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The Information Asymmetry Effects of Expanded Disclosures About Derivative and Hedging Activities

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Abstract. I study the information asymmetry effects of Statement of Financial Accounting Standards Number 161 (SFAS 161), which requires changes to the content and format of derivative and hedging footnote disclosures. Using a difference-in-differences design, I investigate whether these mandatory disclosure changes affected bid-ask spreads. To capture the extent to which firms were likely impacted by SFAS 161, I employ two complementary measures: (1) actual changes in firms' derivative and hedging disclosures, and (2) pre-SFAS 161 levels of firms' derivative and hedging activities. Both measures provide consistent evidence that bid-ask spreads decreased more for firms whose disclosures were more likely affected by SFAS 161. I also find that increased qualitative information and more disaggregated quantitative data (i.e., disclosure content) matter more than disclosure grouping and tabular display (i.e., disclosure format) for the observed decrease in bid-ask spreads. Overall, my findings suggest that the disclosure changes required by SFAS 161 reduced information asymmetry among investors regarding the firm value effects of derivative and hedging activities. These results may prove useful to regulators and standard setters as they consider disclosure requirements in other contexts.

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1. Introduction

In recent years, accounting standard setters and securities regulators have emphasized the need for disclosures that more effectively convey the information that is most relevant to stakeholders (Financial Accounting Standards Board (FASB) 2012, International Accounting Standards Board (IASB) 2013, Securities and Exchange Commission (SEC) 2020). This focus on the effectiveness of regulated financial disclosures highlights the importance of studying the aspects of required disclosures that are, or could potentially be, most useful to stakeholders. However, required disclosures differ in subject, complexity, purpose, and relevance for particular decisions. These differences can pose difficulties for researchers aiming to shed light on the broad topic of disclosure regulation effectiveness. Acknowledging these challenges, Lev (1988) asserts that the existence of disclosure regulation can be justified by the presence of information asymmetries among investors. This useful perspective suggests that "the effectiveness of disclosure regulations" can be determined by their impacts on information asymmetry (Lev 1988, p. 16). The objective of this paper is to provide insight about the effectiveness of disclosure regulation by considering how required changes to derivative and hedging (DH) footnote disclosures imposed by Statement of Financial Accounting Standards Number 161 (SFAS 161) impacted information asymmetry.

For several reasons, I view SFAS 161 as a disclosure regulation deserving focused attention from an information asymmetry standpoint. First, SFAS 161 only impacted disclosures (leaving recognition and measurement unchanged), and the FASB issued it with the express purpose of providing stakeholders with an "enhanced understanding" of firms' DH activities (FASB 2008, paragraph 1). Second, DH disclosures should matter to stakeholders because DH activities themselves are material to firm value, risk exposures, investments, and leverage (e.g., Schrand 1997, Guay 1999, Allayannis and Weston 2001, Pérez-González and Yun 2013, and Manchiraju et al. 2018). Moreover, SFAS 161 was issued as a direct response to complaints about the inadequacy of existing DH information, indicating that (at least some) financial statement users focus on these disclosures and likely paid attention to the changes imposed by SFAS 161. Third, although derivatives and hedging arrangements are certainly not the only complex financial statement components that matter to stakeholders, they are noteworthy because DH activities have become more widespread and increasingly complex over time (FASB 2008, Hull 2012, Bank for International Settlements (BIS) 2018).

Taken together, these reasons underscore the importance of investigating the information asymmetry effects of changes to DH disclosures established by SFAS 161. Because the standard's purpose of improving investors' understanding relates closely to regulators' oft-cited objective of "leveling the playing field" (e.g., Beatty and Hand 1992) and to the concept of relative informedness among investors, Lev's (1988) suggestion to focus on the information asymmetry effects of disclosure regulation seems of particular importance when analyzing the impact of SFAS 161. If the disclosure changes improved all investors' understanding of the firm-value implications of firms' DH activities (as the FASB hoped), investors would be generally better informed, reducing the informational disadvantage of relatively uninformed investors and leading to less information asymmetry. On the other hand, if DH activities are just too complex for most investors to understand, it is possible that only more sophisticated investors would be able to enjoy the benefits of expanded DH disclosures after SFAS 161, increasing these investors' informational advantages and leading to more information asymmetry. In summary, even though SFAS 161 was not issued to specifically impact information asymmetry per se, the direction of any change in asymmetry attributable to SFAS 161 is informative because it provides insight about whether and which investors benefitted from the expanded disclosures. Although my findings must be interpreted in the context of SFAS 161, the evidence may prove useful to regulators and standard setters considering disclosure rules in other contexts (depending on how each context compares to the objectives and requirements of SFAS 161). The results also contribute to the literature on the effectiveness of disclosure regulation, particularly as it relates to information asymmetry among investors.

My empirical investigation relies on a difference-in-differences framework, which requires (1) an outcome measure that represents information asymmetry and (2) treatment measures that capture the extent to which firms' DH footnote disclosures were (likely) impacted by SFAS 161. My outcome variable of interest is the bid-ask spread, which is commonly used to capture information asymmetry among investors (e.g., Bagehot 1971, Copeland and Galai 1983, and Glosten and Milgrom 1985) and is also identified by Lev (1988) as a useful empirical outcome for evaluating disclosure regulation. Despite the conceptual appeal of the bid-ask spread and its prevalence in the literature, using it as the outcome of interest in the SFAS 161 setting relies on two important assumptions.[1] First, because bid-ask spreads relate to asymmetry regarding firm value, I assume DH information is value relevant for investors. DH information is clearly risk relevant because most firms' DH activities are meant to manage risk exposures, but consistent with my assumption, both prior research and the FASB's statements suggest that DH disclosure is also value relevant (e.g., Venkatachalam 1996, Schrand 1997, Ahmed et al. 2006, FASB 2008, and Caban 2018). Second, I assume bid-ask spreads can reflect asymmetry arising from informational differences about how firms' risk exposures and related DH activities impact firm value. This assumption seems reasonable because differences in individuals' interpretations of public, value-relevant information can lead to information asymmetry, particularly with complicated information (e.g., Kim and Verrecchia 1994 and Linsmeier et al. 2002). DH disclosures appear to fit this scenario given the inherent complexity of DH activities and financial statement user complaints about pre-SFAS 161 DH disclosures (FASB 2008).[2]

Turning to the treatment aspect of my difference-in-differences design, I employ two complementary measures. First, to directly capture the intended effect of SFAS 161, I use textual analysis to measure *changes in* the content and format of firms' 10-K DH footnote disclosures following the adoption of SFAS 161.[3] Because SFAS 161 required new details and changed the structure of DH disclosures, I measure the amount of qualitative DH information by calculating the percentage of all financial statement footnote words related to derivatives or hedging. This percentage is based on a dictionary of 231 words, phrases, and acronyms that significantly expands the words and phrases used in prior literature to locate DH disclosure in firms' SEC filings. Next, because SFAS 161 required more disaggregated fair values, gains, and losses, I measure the amount of quantitative DH information by counting numbers appearing in DH sentences and tables. These first two characteristics are proxies for the content of DH footnote disclosures. Because SFAS 161 required DH disclosures to be presented in one footnote or a series of cross-referenced footnotes, I also quantify the extent to which qualitative DH information is grouped together in the financial statement footnotes. Finally, I measure the percentage of total quantitative DH information appearing in tables as SFAS 161 imposed tabular presentation requirements. These last two characteristics are proxies for the format of DH footnote disclosures. I use factor analysis to combine *changes in* these four disclosure variables into a summary measure of DH footnote disclosure change for each firm (*DHDISC*). This treatment measure is useful because it represents variation in the impact of SFAS 161 across firms and directly captures the FASB's intentions in requiring firms to change their DH disclosures. However, despite its intuitive appeal, *DHDISC*'s drawbacks should not be ignored. For example, *DHDISC* necessarily relies on postadoption data to calculate DH disclosure changes. This aspect of the treatment measure is problematic for drawing causal inferences because *DHDISC* is not determined before adoption and is itself directly affected by firms implementing SFAS 161 (e.g., Armstrong and Kepler 2018). Moreover, other factors (besides SFAS 161) could possibly influence changes in firms' DH disclosures concurrent with the adoption of SFAS 161, leading to potentially biased estimates of SFAS 161's impact on information asymmetry when using *DHDISC* as the treatment measure.

To ameliorate these concerns, I create a second treatment measure using DH data from Compustat in the pre-SFAS 161 period to proxy for the extent of firms' DH activities leading up to the adoption of the new disclosure requirements. More specifically, I use the absolute value of derivative-related gains or losses in comprehensive income (scaled by assets), averaged over the eight quarters ending one year prior to the adoption of SFAS 161 (*DHACTIVITY*). This treatment measure uses only preadoption data and thus addresses the primary concerns with *DHDISC*; however, *DHACTIVITY* also has weaknesses. For example, it does not capture all types of DH activity because only certain types of DH arrangements (primarily designated cash flow hedges) are reflected as gains or losses in comprehensive income.[4] Moreover, although it is intuitive that high-*DHACTIVITY* firms should be more strongly impacted by SFAS 161 because their DH disclosures should change more, *DHACTIVITY* cannot account for the possibility that some firms provide better DH disclosures prior to SFAS 161. Last, firms with higher versus lower *DHACTIVITY* are likely different in fundamental ways, implying that any observed difference in bid-ask spreads around SFAS 161 adoption could be attributable to other factors besides the new DH disclosure requirements. To summarize, *DHDISC* and *DHACTIVITY* have different advantages and disadvantages from a research design perspective. These differences make it unlikely that any single factor would simultaneously confound inferences based on both measures; thus, consistent evidence across both treatment measures would help rule out alternative (i.e., non-SFAS 161) drivers of the results.

Using these measures with my difference-in-differences framework, I find that firms whose DH disclosures likely changed more as a result of adopting SFAS 161 experienced significant decreases in bid-ask spreads. Importantly, the results are very similar when using either *DHDISC* or *DHACTIVITY* as the measure of treatment, lending credibility to the findings. These results are consistent with SFAS 161 reducing information asymmetry among investors, suggesting that investors generally increased their understanding of how firms' DH activities impact firm value, rather than only expert investors enhancing their informational advantages. Because SFAS 161 became effective during the 2008 financial crisis, I make several research design choices to alleviate concerns that my results are potentially driven by (1) the differential impact of the crisis on DH versus non-DH firms or (2) material changes in firms' DH and/or risk management behaviors. To address the first concern, using *DHDISC* and *DHACTIVITY* in my difference-in-differences models allows me to employ "a continuum of possible treatments" (Wooldridge 2010, p. 150), which can more precisely capture the firm-specific impact of SFAS 161 rather than just comparing affected firms (i.e., DH firms) to nonaffected firms (i.e., non-DH firms). Because the treatment measures are continuous, I am able to estimate the effect of SFAS 161 on bid-ask spreads based on variation in the extent of treatment *only among DH firms*, and I find very similar results. I also conduct parallel trends analyses and provide evidence suggesting that preadoption trends in bid-ask spreads were similar for both treatment and control firms when using either *DHDISC* or *DHACTIVITY* as the treatment measure. Additional results based on pseudo dates for adopting SFAS 161 also bolster the main findings. To address the second concern, I show that DH and non-DH firms are not different in terms of *changes in* overall risk exposure during the sample period, suggesting that DH firms did not drastically change their risk management activities around SFAS 161. I also find that the results are robust to the inclusion of several controls for changes in firms' risk management and/or DH activities (Chiorean 2016, Calluzzo and Dudley 2018).

Last, I address the secondary objective of this paper by using the four components of *DHDISC* that capture DH disclosure content versus format to investigate whether specific aspects of SFAS 161 had more impact than others on bid-ask spreads. Using these variables as alternative treatment measures in my difference-in-differences models, I find that the effect of SFAS 161 on bid-ask spreads is driven primarily by increased qualitative and disaggregated quantitative DH information. These findings suggest that the required content changes in DH footnote disclosures following SFAS 161 were more effective than required format changes in reducing information asymmetry among investors.

Altogether, my findings are consistent with SFAS 161 disclosures reducing information asymmetry as captured by the bid-ask spread. I base this inference on consistent results across two treatment measures with contrasting (dis)advantages, along with several supporting robustness tests. Nevertheless, my conclusions are subject to the important proviso that I cannot completely rule out the possibility that some unidentified factor is responsible for the documented decrease in bid-ask spreads. Moreover, whereas the results may prove useful to regulators and standard setters as they continue to deliberate the effectiveness of financial statement disclosures, the evidence presented here should be interpreted in the specific context of SFAS 161. For example, my results suggest that the DH disclosure changes required by SFAS 161 led to better understanding of DH activities for investors generally rather than solely benefitting the more informed subset of investors. Similarly, I provide evidence that increased amounts of qualitative information and more disaggregated quantitative information (i.e., DH disclosure content) had stronger impacts on information asymmetry than did changes to the grouping or tabular formatting of information (i.e., DH disclosure format). However, it remains unclear whether these effects would translate to disclosures about other financial statement components.

This paper makes several contributions to the accounting literature. Because DH disclosures are complex and difficult to analyze, most research in this area uses hand-collected data to study relatively small samples or certain industries. Although this approach is useful, it is important to consider the effects of DH disclosures for all firms. By creating a new dictionary of DH words and phrases and using textual analysis, I am able to study a broad sample of over 2,000 firms from all industries, measuring with greater precision the content of DH disclosure in financial reports. I also develop methods to identify the format of DH information. Whereas prior research on such disclosure attributes tends to focus on one attribute at a time, my approach enables the simultaneous study of several disclosure characteristics, allowing for a more nuanced view of which disclosure characteristics matter more to financial statement users. Moreover, these techniques could be easily adapted to study the content and format of disclosures for any financial statement component or topic, providing a useful framework for examining additional disclosure regulations.

2. Background, Prior Literature, and Research Questions

2.1. SFAS 161 Disclosures

Before SFAS 161, DH footnote disclosures were governed by SFAS 133 and its amendments and interpretations.[5] On the basis of feedback from investors and other financial statement users, the FASB determined that the required disclosures were insufficient for comprehending the firm-value effects of firms' DH activities (FASB 2008). As a result, the board issued SFAS 161 to amend SFAS 133 with the purpose of providing financial statement users with an "enhanced understanding" of how firms' DH activities "affect an entity's financial position, financial performance, and cash flows" (FASB 2008, paragraph 1). SFAS 161 required new disclosures, the disaggregation of quantitative information, and changes to several characteristics of the required disclosures; I describe these changes in the subsections that follow. Appendix A summarizes the SFAS 133 and SFAS 161 disclosure requirements, and Figure IA.1 in the internet appendix shows derivative disclosures under both standards for the Perrigo Company.[6]

2.1.1. New Qualitative Disclosures and Structure. SFAS 161 changed the structure of firms' narrative disclosures about DH objectives, from organization by accounting hedge designation (SFAS 133) to organization by underlying risk exposure. SFAS 161 also required new disclosures about credit-risk-related contingent features, the events that would trigger the contingencies, and the fair values of any net-liability derivatives containing credit-risk features. Firms must also disclose fair values of assets that would be used as collateral or to settle the derivative instruments if credit-risk contingencies are triggered. SFAS 161 also required firms to provide information about the volume of their derivative activity and to indicate which line items on the balance sheet (income statement) contain DH fair values (gains and losses).

2.1.2. Disaggregated Quantitative Disclosures. Instead of requiring separate fair values only for assets and liabilities (SFAS 133), SFAS 161 mandated that DH fair values be presented separately for assets, liabilities, hedging instruments, nonhedging instruments, and each risk exposure category.[7] Similarly, SFAS 161 increased disaggregation for gains and losses by requiring the disclosure of more than twice the number of separate gains and losses, presented separately for each underlying risk exposure category.

2.1.3. Disclosure Frequency, Display, Grouping, and Flexibility. SFAS 161 increased the frequency of DH footnote disclosures, from only annual periods to both interim and annual reporting periods. After considering the cost of preparing quarterly disclosures, the board concluded that financial statement users would benefit as a result of "frequent and often significant changes in derivative fair values" (FASB 2008, paragraph A50). Firms must also use tabular presentation for most quantitative DH information, and they have to cross-reference their disclosures if DH information is presented in more than one footnote. Finally, SFAS 161 provided flexibility to preparers in how to disclose the volume of derivative activity. For example, preparers might disclose notional amounts or the quantity of commodities purchased forward.

2.2. Prior Literature

My study of SFAS 161 disclosures builds on and contributes to three areas of prior research: (1) investor perception of DH activities, (2) financial reporting and disclosure for DH activities, and (3) effects of disclosure characteristics such as presentation format and information grouping.

2.2.1. Investor Perception of Derivative Use and Hedging Activities. Theoretical research on risk management and hedging identifies several factors that influence the decision to hedge. Examples include private information, career concerns, risk aversion, and/or price uncertainty (e.g., Holthausen 1979; Stulz 1984; and DeMarzo and Duffie 1991, 1995). Similarly, empirical studies find that both managerial characteristics and firm characteristics are associated with DH activities (e.g., Tufano 1996 and Géczy et al. 1997). Because many factors affect the decision to hedge and/or use derivatives, it is important to consider how investors view these activities. Koonce et al. (2008) use an experiment to study how investors perceive managers using derivatives. They find that investors favorably perceive managers using nonspeculative derivatives to manage risks, suggesting that investors are concerned about "how and why an entity uses derivative[s]" (FASB 2008, paragraph 1). In a separate experiment, Koonce et al. (2005) show that different descriptions of economically equivalent derivatives and other financial instruments cause investors to make different risk assessments. In light of the aforementioned evidence that managers' DH decisions are influenced by diverse factors, have material effects on firms, and matter to investors, the findings by Koonce et al. (2005) emphasize the importance of studying how financial statement users are influenced by DH *disclosures*. I extend this literature by using a large archival sample to study how information asymmetry is affected by changes to the content and format of required footnote disclosures for DH activities.

2.2.2. Financial Reporting and Disclosure for Derivatives and Hedging. Accounting standard setters have periodically reconsidered financial reporting and disclosure for DH activities for nearly three decades, and researchers have studied the usefulness of these standards from several perspectives. The weight of the evidence from this research is that DH disclosures are relevant for evaluating firm risk (e.g., McAnally 1996 and Ahmed et al. 2011), firm value (e.g., Venkatachalam 1996, Schrand 1997, Ahmed et al. 2006, and Caban 2018), firms' risk exposures (e.g., Linsmeier et al. 2002 and Zou 2019), and the impact of choosing to use hedge accounting (e.g., Zhang 2009, Panaretou et al. 2013, Cowins 2014, Chiorean 2016, and Pierce 2020).[8]

The findings from these studies suggest that DH disclosures are useful to market participants, and I extend this literature in two ways. First, because of the complex nature of these disclosures and the reliance on hand-collected data, the extant literature tends to rely on relatively small samples (typically between 150 and 250 firms) and/or focuses on specific industries.[9] Using textual analysis methods allows me to examine DH disclosures for a much larger sample of firms spanning many industries. This approach extends the literature by providing a more comprehensive view of the impact of DH disclosures and also directly considers the FASB's decision to require expanded DH disclosures for all firms and not just those with extensive DH activities. Second, my investigation of how SFAS 161 disclosures affect information asymmetry extends the literature by shedding more direct light on the FASB's stated objective for SFAS 161: improving financial statement users' understanding of firms' DH activities. This perspective contributes to the SFAS 161 literature that so far has focused on competitors' use of disclosure (Zou 2019), the impact of DH customers on suppliers (Chen et al. 2021), value relevance (Caban 2018), the effects of hedge accounting, the implications of cash flow hedges, and whether firms use derivatives to hedge or speculate (Cowins 2014, Chiorean 2016, Manchiraju et al. 2018, Pierce 2020, Campbell et al. 2021).

2.2.3. Disclosure Characteristics. My study of SFAS 161 also relates to research on the effects of presentation format, information grouping, and the interplay between qualitative and quantitative information. Mentioned previously, Koonce et al. (2005) show that investors' risk assessments are affected by the way derivatives and financial instruments are described. Prior experimental research also finds that presentation format influences financial statement users' information acquisition (Hirst and Hopkins 1998), information weighting (Maines and McDaniel 2000), and processing fluency (Rennekamp 2012). Using an archival sample, Linsmeier et al. (2002) find that the effectiveness of market risk disclosure formats depends on the type of risk being evaluated. Bloomfield et al. (2015) use an experiment to show that credit analysts' ability to identify relevant information in financial reports is improved when similar information is grouped in the same section of a financial disclosure. The authors argue that coordinated presentation "reduc[es] the cognitive load necessary for integrating the related information and forming a meaningful mental model of the firm" (p. 525). Relatedly, Allee and DeAngelis (2015) find that the dispersion of tone words in conference calls is informative about performance, concluding that "tone dispersion both reflects and affects the information that managers convey through their narratives" (p. 243).

The findings from these papers emphasize that the way information is presented has a measurable impact, highlighting the importance of considering the different DH disclosure characteristics affected by SFAS 161. I extend and complement prior studies by applying textual analysis to a broad archival sample of 10-K footnotes and by *simultaneously* studying both disclosure *content* (qualitative and quantitative DH information) and disclosure *format* (tabular display and grouping of DH disclosures within financial statement footnotes). The *New Oxford American Dictionary* (3rd ed.; see Endnote 3) defines *content* as "the substance or material dealt with in a speech, literary work, etc., as distinct from its form or style" and defines *format* as "the way in which something is arranged or set out." The SFAS 161 Basis for Conclusions (see paragraphs A1–A77 of FASB 2008) indicates that the FASB viewed the required changes to both content and format as improvements to DH disclosures, but the board did not indicate which changes (i.e., content versus format) were expected to matter most for improving investor understanding. Although content and format are clearly related and cannot be completely distinguished (i.e., something can only be formatted in a particular way if the content exists), it is important to consider each as a specific "tool" at regulators' disposal when they set disclosure requirements. Relative to focusing on one particular characteristic, a multipronged analysis permits inferences about which characteristic(s) matter more to financial statement users and thus which characteristics might warrant particular consideration by regulators.

2.3. Research Questions

My primary research question considers whether the changes to DH footnote disclosures required by SFAS 161 affected information asymmetry among investors, and my secondary question focuses on which disclosure characteristics (i.e., content versus format) drive this potential impact of SFAS 161. I focus on information asymmetry because prior research shows that it can be influenced by information disclosed in financial reports (e.g., Diamond 1985, Kim and Verrecchia 1994, Shroff et al. 2013, and Bens et al. 2016). I also rely on Lev's (1988) perspective in choosing to study the information asymmetry effects of SFAS 161. Lev (1988) argues that disclosure regulation can be justified by a desire to reduce information asymmetries among investors because uninformed investors' protective actions (i.e., when trading with informed parties) lead to undesirable capital market outcomes such as high transactions costs and lower liquidity. A direct corollary of this argument is that information asymmetry provides a useful lens for examining the effectiveness of disclosure regulations. This perspective is consistent with regulators' oft-mentioned goal of "leveling the playing field" among investors (e.g., Beatty and Hand 1992) and seems especially suitable for evaluating the effects of SFAS 161 because its stated goal of improving investors' understanding is closely related to relative informedness among investors. In summary, if a shift in DH disclosure requirements impacts investors' understanding of how firms are affected by DH activities, this impact should correspond to observable changes in information asymmetry.[10]

The FASB clearly hoped that issuing SFAS 161 would lead to an improved understanding of DH activities for all financial statement users; however, the direction of SFAS 161's eventual impact on information asymmetry is less clear because of uncertainty about (1) whether overall investor understanding would improve and (2) which investors' understanding might improve most. For example, whereas the stated dissatisfaction with pre-SFAS 161 DH disclosures suggests that the required changes would be viewed as an improvement, comment letters on the SFAS 161 exposure draft indicate mixed opinions about how the proposed DH disclosure changes might affect financial statement users. Table IA.1 in the internet appendix provides excerpts from several letters indicating the range of viewpoints on the issue. Whereas some users strongly supported the FASB's efforts, others were concerned about stakeholders becoming *more* confused as a result of the proposed disclosure changes. Moreover, the FASB recently issued ASU 2017-12, "Targeted Improvements to Accounting for Hedging Activities," to address hedging in the context of risk management (FASB 2017). Although this additional regulation could reflect overtime increases in the complexity of DH arrangements, it could also indicate that SFAS 161 did not fully accomplish the board's objective of enhancing investor understanding. Consistent with the latter, FASB member Jim Kroeker commented on the need for new disclosures related to risk management (i.e., beyond those required by SFAS 161): "It's hard to get an appreciation from the financial statements today [i.e., after implementing SFAS 161] of what percentage of the respective exposure is actually being hedged.… You really don't get a sense for what was the entity's goal, what was its objective, and then how successful was it at accomplishing that objective."[11]

Furthermore, it is possible that DH subject matter is just too complex for a disclosure improvement to meaningfully improve understanding for most investors. Chang et al. (2016) study how analyst forecasts are affected when firms begin using derivatives. They find evidence that "analysts routinely misjudge the earnings implications of firms' derivatives activity" because of the financial reporting complexities associated with derivative use (p. 584). Although they conclude that over-time improvements to derivative standards have helped analysts, Chang et al. (2016) do not find that SFAS 161 remedies the negative effects of DH activities on forecast accuracy and dispersion. Although focused on earnings forecasts, this evidence suggests that DH activities are complex enough that even experienced financial statement users may not fully understand their effects.[12] Another implication of this perspective is that because DH activities are complex, the returns to costly information processing are high (Kim and Verrecchia 1994, Linsmeier et al. 2002), and a small number of expert investors could benefit from increased DH disclosures while the vast majority of investors may not. If this were the outcome, then SFAS 161 would result in greater informational advantages for a subset of investors rather than putting all investors on a more level playing field in terms of understanding the firm-value effects of firms' DH activities.

Taken together, the arguments discussed in the preceding paragraphs indicate mixed views about the direction of the potential impact SFAS 161 may have had on information asymmetry among investors.[13] If most investors were able to improve their understanding of firms' DH activities, investors would be generally more informed, reducing the informational disadvantage of relatively uninformed investors and decreasing information asymmetry. On the other hand, if SFAS 161 disclosures create more confusion about how firms' DH activities affect firm value, or if the new disclosures benefit only a select few expert investors, then information asymmetry could increase after the adoption of SFAS 161. I therefore turn to my empirical analyses to investigate how SFAS 161 affected information asymmetry among investors.

3. Research Design and Sample

3.1. Research Design

To evaluate the impact of SFAS 161-imposed disclosure changes on information asymmetry, I use a sample of firm-quarter observations centered on the adoption of SFAS 161 with a difference-in-differences framework including firm and time fixed effects. I employ the following baseline model:

$$SPREAD_{it} = \beta_0 + \beta_1 POST_{it} \times TREAT_i + \beta_2 POST_{it} + \sum_{k=1}^{k=n} \beta_{3,k} CONTROL_{k,it} + \gamma_i FIRM_i + \alpha_t TIME_t + \varepsilon_{it}. \tag{1}$$

In Equation (1), $i(t)$ is the firm (quarter) index, and *SPREAD* is my measure of information asymmetry based on bid-ask spreads. *POST* is an indicator variable equal to 1 (0) for all firm-quarters with 10-Q or 10-K filing dates on or after (before) the firm's initial SFAS 161 filing, and *TREAT* represents a firm-specific variable capturing the extent to which the firm was (likely) impacted by SFAS 161. The interaction term $POST \times TREAT$ is the variable of interest and captures the impact of SFAS 161 on *SPREAD*.[14] The following subsections describe the components of Equation (1) in more detail.

3.1.1. Measuring Information Asymmetry Using the Bid-Ask Spread. I focus on the bid-ask spread because of its long history as a measure of information asymmetry about firm value among investors (e.g., Bagehot 1971, Copeland and Galai 1983, and Glosten and Milgrom 1985). Lev (1988) also identifies the bid-ask spread as a useful outcome measure for evaluating disclosure regulation. As mentioned in the introduction, using the bid-ask spread for assessing the impact of SFAS 161 relies on two key assumptions. First, DH information is value relevant for investors. Given that DH activities are closely related to managing risk exposures, DH information is clearly risk relevant (e.g., McAnally 1996, Linsmeier et al. 2002, Ahmed et al. 2011, and Zou 2019). DH information is also likely value relevant because it helps investors form expectations about how fluctuating risk factors affect firm value over time. Consistent with this view, the FASB views DH disclosures as important for understanding how DH activities "affect an entity's financial position, financial performance, and cash flows," all of which are directly related to firm value (FASB 2008, paragraph 1). As already mentioned, prior research also concludes that DH information is value relevant (e.g., Venkatachalam 1996, Schrand 1997, Ahmed et al. 2006, and Caban 2018).

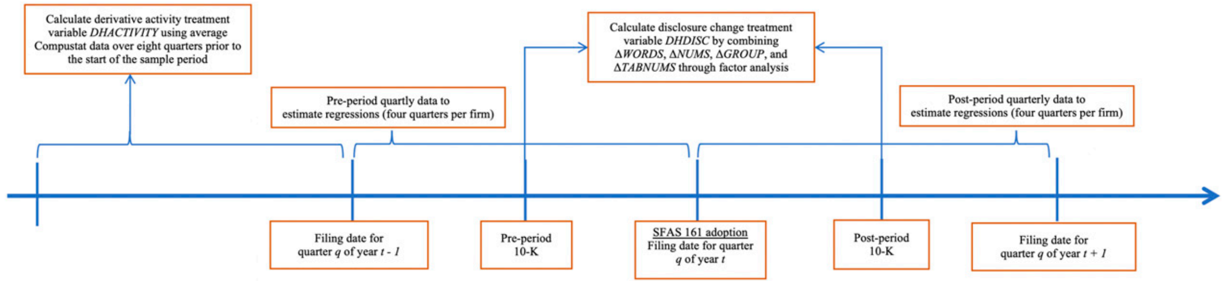
Second, bid-ask spreads can reflect informational asymmetries arising from informational differences about how firms' risk exposures and related DH activities impact firm value.[15] For example, if oil is an important input for a firm, I assume that asymmetry can arise among investors from informational differences about how fluctuations in oil prices impact firm value in light of the firm's DH activities. Kim and Verrecchia (1994) suggest that bid-ask spreads can increase when disclosures "stimulate informed judgments among traders who process public disclosure into private information" (p. 44), and Linsmeier et al. (2002) apply these ideas to the setting of firms' risk exposures, arguing that "movements in underlying market rates … have firm-specific implications because they are associated with equity returns and earnings" (p. 349). Linsmeier et al. also contend that lower-quality risk disclosures can increase the perceived returns to investors' efforts to learn about firms' exposures and their impact on firm value. Given that DH activities are intrinsically complex and that financial statement users were dissatisfied with existing (i.e., pre-SFAS 161) DH disclosures, it seems plausible that investors' efforts to learn about firms' risk exposures and DH activities could lead to more information asymmetry and larger spreads.

To construct *SPREAD* for estimating Equation (1), I follow Bens et al. (2016) and focus on the month following the 10-K/10-Q filing date pertaining to each firm-quarter. Bens et al. only include 10-K filing dates in their analyses because they are focused on fair value estimates, and they argue that the 10-K contains "the most comprehensive set of information about fair value estimates and disclosure" (p. 354).[16] However, SFAS 161 requires the same DH disclosures for annual and interim periods, so I include 10-Q filing dates in addition to 10-K filing dates and define *SPREAD* for each firm-quarter as the natural log of the average daily bid-ask spread during the month beginning on the corresponding 10-Q/K filing date.[17]

3.1.2. Treatment Measures: DH Footnote Disclosure Changes and Preadoption DH Activity. In Equation (1), *TREAT* can be one of two continuous, firm-level measures (that do not vary with time during the sample period) meant to capture the extent to which each firm is (likely) impacted by SFAS 161. I use automated textual analysis to create the first measure, which relates directly to the implementation of SFAS 161 because it captures actual changes in the content and format of firms' DH footnote disclosures. Because SFAS 161 changed qualitative DH disclosures, I start by measuring the amount of qualitative DH information as the percentage of all words in the financial statement footnotes related to DH activities (*WORDS*). Because SFAS 161 required more disaggregated numerical information, I next measure the amount of quantitative DH information as the count of numbers appearing in DH sentences and tables, multiplied by 100 and scaled by the total number of words in the financial statement footnotes (*NUMS*). SFAS 161 also required DH disclosures to be presented in either one footnote or a series of cross-referenced footnotes, so I measure the extent to which qualitative DH information is grouped together in the footnotes.[18] Following Allee and DeAngelis (2015), *GROUP* is defined as the average reduced frequency (ARF) of DH words, phrases, and acronyms in the footnotes. Finally, because SFAS 161 requires tabular display for quantitative information, I measure the percentage of total quantitative DH information appearing in tables. *TABNUMS* is defined as the count of DH numbers presented in tables (one of the components of *NUMS* described previously) multiplied by 100 and divided by the total count of DH numbers.

Changes in these four measures capture separate DH disclosure effects of SFAS 161. More specifically, I measure the change in each DH disclosure characteristic as the difference between (1) the characteristic from the firm's first 10-K filing footnotes under SFAS 161 and (2) the characteristic from its last 10-K filing footnotes under SFAS 133 (see Figure 1).[19] This process results in the following disclosure change measures: Δ*WORDS*, Δ*NUMS*, Δ*GROUP*, and Δ*TABNUMS*.[20] The first two (Δ*WORDS* and Δ*NUMS*) capture changes in DH disclosure *content*, whereas the last two (Δ*GROUP* and Δ*TABNUMS*) capture changes in DH disclosure *format*, which I analyze separately in Section 5. To construct my first *TREAT* measure for estimating Equation (1), I combine Δ*WORDS*, Δ*NUMS*, Δ*GROUP*, and Δ*TABNUMS* through factor analysis. The resulting factor variable is *DHDISC*, which captures the overall change to DH footnote disclosures caused by SFAS 161.[21] *DHDISC* is an appealing measure of the extent to

Figure 1. (Color online) Research Design



Notes. This figure summarizes the research design for estimating Equation (1). SFAS 161 adoption periods are either (1) a first, second, or third fiscal quarter ending between February 15, 2009, and May 15, 2009; or (2) a first fiscal quarter ending between May 16, 2009, and August 15, 2009. The pre- and postperiods are each defined to contain four fiscal quarters. Observations are retained for all 10-Q and 10-K filings between the filing date for quarter q of year $t-1$ (inclusive) and the filing date for quarter q of year $t+1$ (not inclusive). Sample firms are required to have eight nonmissing quarterly filings in this time period and nonmissing regression variables for all eight quarters.

which firms are impacted by SFAS 161 because it directly captures the disclosure changes required by the FASB and allows for variation across firms. However, *DHDISC*'s reliance on postadoption disclosures is a weakness. This aspect of the measure is problematic for drawing causal inferences about the impact of SFAS 161 because *DHDISC* is not determined before adoption and is directly affected by firms implementing SFAS 161 (e.g., Armstrong and Kepler 2018). Furthermore, if anything besides SFAS 161 affected the changes in firms' DH disclosures captured by *DHDISC*, the estimated treatment effects will be biased.

To address these disadvantages, I create a second firm-specific *TREAT* measure that captures variation in the level of firms' DH activities before the adoption of SFAS 161. More specifically, using Compustat data over the eight quarters ending one year before the adoption of SFAS 161 (see Figure 1), I calculate the average of the absolute value of derivative-related gains or losses in comprehensive income (scaled by total assets): *DHACTIVITY*. The intuition for this approach is that firms with greater levels of *DHACTIVITY* have more extensive DH activities and are those whose disclosures are likely to be materially impacted by adopting SFAS 161. Because *DHACTIVITY* does not require posttreatment data, it avoids the primary downside of *DHDISC*, but it also has its own drawbacks. First, because of Compustat's limited DH data in the pre-SFAS 161 period, not all DH arrangements are captured by *DHACTIVITY*. Only certain DH arrangements (primarily those designated as cash flow hedges) impact comprehensive income; thus, firms with a low *DHACTIVITY* measure may, in fact, have extensive DH activities. Second, unlike *DHDISC*, *DHACTIVITY* ignores firms potentially providing better DH disclosures before SFAS 161. For example, in its SFAS 161 comment letter, National City Corporation stated, "Years ago, we decided that our financial statement users needed more disclosure on our use of derivatives. Thus, our current disclosures go beyond what is currently required" (Kelly 2007). Such firms may have high values of *DHACTIVITY*, but they would not be affected by SFAS 161 as much as firms with similar DH activities that were disclosing the bare minimum.[22] Finally, firms with higher versus lower *DHACTIVITY* likely differ in important ways, implying that any observed difference in *SPREAD* around SFAS 161 adoption could be attributable to other factors besides the new DH disclosure requirements. In the end, *DHDISC* and *DHACTIVITY* have different merits and shortcomings from a research design perspective. As a result, it is unlikely that any single factor would simultaneously confound inferences based on both measures. I thus present results and draw conclusions based on both treatment measures in order to minimize the likelihood of drawing spurious inferences.

Importantly, both *DHDISC* and *DHACTIVITY* are firm-specific, continuous variables. This method modifies the traditional difference-in-differences model to accommodate "a continuum of possible treatments" (Wooldridge 2010, p. 150) and is important for two reasons. First, continuous treatment measures allow for cross-sectional variation in the treatment effect on *SPREAD*. Thus, instead of drawing inferences by simply comparing firms with DH activities (those adopting SFAS 161) to non-DH firms (those unaffected by SFAS 161), inferences can be made based on the extent to which SFAS 161 likely impacted firms' DH disclosures. Consistent with Schwert's (1981) argument that many regulations affect individual firms differently, any shift in *SPREAD* induced by SFAS 161 should be greater for firms whose disclosures changed more, but a binary *TREAT* measure cannot account for such variation. Second, the continuous treatment approach allows me to reestimate the difference-in-differences models using *only firms with DH activities*. This ability is particularly important in my setting because firms adopted SFAS 161 during the 2008 financial crisis, and the decision to engage in DH activities is not random. Thus, bid-ask spreads may behave differently for DH and non-DH firms during the sample period for reasons other than SFAS 161 disclosure changes.[23] Restricting the analysis to DH firms maintains the rigor of the difference-in-differences framework with firm fixed effects but also provides additional assurance that results are, in fact, driven by SFAS 161 disclosure changes and not by other unrelated factors that differ between DH and non-DH firms.

To summarize, *TREAT* in Equation (1) can be *DHDISC* or *DHACTIVITY*. Negative (positive) estimates of the *POST* × *TREAT* coefficient in Equation (1) would suggest that DH disclosure changes imposed by SFAS 161 reduced (increased) information asymmetry among investors. Moreover, if SFAS 161 is driving any observed impact on *SPREAD*, I would expect similar *POST* × *TREAT* coefficients whether using *DHDISC* or *DHACTIVITY* as the treatment measure and whether using the full or DH firm sample.

3.1.3. Control Variables. Since SFAS 161 became effective near the end of the 2008 financial crisis, time (*TIME*) and firm (*FIRM*) fixed effects in the difference-in-differences framework of Equation (1) are critical for identifying any information asymmetry effects driven by the new SFAS 161 disclosures. The time fixed effects are based on the calendar year-quarter of the relevant 10-Q or 10-K filing date and control for marketwide fluctuations during the sample period; the firm fixed effects capture firm characteristics not measured by the n firm-quarter controls in Equation (1). Moreover, the sample period is relatively short (restricted to four preadoption quarters and four postadoption quarters per firm—see Figure 1), which means the firm fixed effects control for firm characteristics *during this sample period* and not firm characteristics present over several years or decades. As a result, the firm fixed effects should capture any firm-specific impact of the financial crisis during the sample period, thereby improving the model's ability to identify the information asymmetry effects of SFAS 161 disclosure changes. The *CONTROL* variables included in Equation (1) follow Bens et al. (2016) and are defined in Appendix B: return on assets (*ROA*), leverage (*LEV*), book-to-market ratio (*BTM*), size (*SIZE*), average stock price (*PRICE*), average trading volume (*TURNOVER*), analyst following (*FOLLOW*), previous return volatility (*PASTSTDRET*), and average trade size (*TRADESIZE*). I also include the average value of the Chicago Board Options Exchange (CBOE) Volatility Index (*VIX*) and the average fluctuations in the firm's underlying market risk factors (*DRISK*). As mentioned in the preceding, one of my important assumptions is that bid-ask spreads can reflect information asymmetry about how firms' risk exposures impact firm value. Thus, although *DRISK* is not the main variable of interest in Equation (1), a significantly positive *DRISK* coefficient would support this assumption.

3.2. Sample

Panel A of Table 1 describes my sample, which is generated by the intersection of the Wharton Research Data Services (WRDS) SEC Analytics Suite, Compustat, and Center for Research in Security Prices (CRSP) databases. The WRDS SEC Analytics Suite is used to identify the filing dates of firms' initial SFAS 161 filings. Because SFAS 161 is effective for all fiscal years and interim reporting periods beginning after November 15, 2008, I identify adoption periods as either (1) a first, second, or third fiscal quarter ending between February 15, 2009, and May 15, 2009; or (2) a first fiscal quarter ending between May 16, 2009, and August 15, 2009.[24] Sample firms are required to have Compustat and CRSP data, along with valid disclosure variables calculated from their pre- and postadoption 10-K forms (see Figure 1). These criteria do not exclude firms for any reason other than data availability, resulting in the maximum number of treatment and nontreatment firms in all industries. The final sample consists of 2,079 firms.

Panel B of Table 1 presents the Fama–French 12 industry distribution of the sample firms separated by SFAS 161 treatment. The sample contains 1,328 (64%) DH firms (observations with *DHFIRM* = 1) and 751 (36%) non-DH firms (observations with *DHFIRM* = 0).[25] All 12 industries are represented in the DH and non-DH groups, but the industry composition differs across the groups (Pearson's $\chi^2 = 128.5$, $p < 0.01$, untabulated). However, three of the four most frequent industries for DH firms are also three of the four most frequent industries for non-DH firms, and these three industries account for 52% of all sample firms.[26] Despite overall differences in industry composition, both DH and non-DH firms are represented in these three common industries, decreasing the likelihood that my results are driven by industry-specific factors.

4. Empirical Results

4.1. Treatment Measures

Table 2 presents summary statistics for the two treatment variables (*DHDISC* and *DHACTIVITY*) along with the DH disclosure measures used to generate *DHDISC* (Δ*WORDS*, Δ*NUMS*, Δ*GROUP*, and Δ*TABNUMS*). Descriptive statistics are presented for DH and non-DH firms, along with corresponding tests for differences in means and medians. If my classification of DH versus non-DH firms is valid, and if my treatment measures indeed capture the extent to which firms are likely affected by SFAS 161, DH firms should exhibit higher values of *DHACTIVITY* and higher values of each of the footnote-based disclosure change measures (compared with non-DH firms). As expected, relative to non-DH firms, DH firms' post-SFAS 161 footnotes contain significantly more quantitative and qualitative DH information, group DH information more closely, and use significantly more tabular display. In addition, the mean and median values of *DHDISC* and *DHACTIVITY* are significantly higher for DH firms compared with non-DH firms.[27]

4.2. Descriptive Statistics

Table 3 presents descriptive statistics for the variables used to estimate Equation (1). Panel A provides statistics for the full sample, whereas Panel B presents statistics separately for DH and non-DH firms. Panel B indicates significant differences in means and medians for nearly all of the all firm-specific variables, confirming expectations that DH firms are generally different from non-DH firms. For the variables in common with Bens et al. (2016) (all variables in Table 3 except for *POST*, *DRISK*, and *VIX*), the average values of the variables in Panel A are similar to those in Table 1 of Bens et al. (2016), although some differences are expected, given that I use quarterly data around the issuance of SFAS 161 and Bens et al. (2016) use annual data over a longer sample period (2007–2012).[28] By construction, half the observations are from the post-SFAS 161 period; consistent with Table 1, 64% of the observations are from DH firms.

4.3. Main Results

Tables 4–8 present results for my primary research question of whether, and in which direction, SFAS 161 disclosure changes affect information asymmetry

Table 1. Sample Description

Panel A: Sample selection procedures

1. Firms with WRDS SEC Analytics Suite filings 10-K, 10-KNT, 10-Q, 10-QT, 10KSB, or 10QSB with the following characteristics: (1) filed between January 1, 2007, and December 31, 2010; (2) nonmissing *GVKEY*, fiscal period end date (*RDATE*), and filing date (*FDATE*); (3) *FDATE* after *RDATE*; and (4) no duplicates on *GVKEY-RDATE.*	9,610
2. Firms without Compustat data or with duplicate Compustat data.	(1,947)
3. Firms headquartered outside the United States.	(485)
4. Firms with filing date errors (absolute difference between *FDATE* year and Compustat fiscal year greater than 1).	(15)
5. Firms with filings not associated with initial adoption of SFAS 161. SFAS 161 adoption observations meet one of two conditions: (1) a first, second, or third fiscal quarter ending between 2/15/2009 and 5/15/2009; or (2) a first fiscal quarter ending between 5/16/2009 and 8/15/2009. Firms also must have a preperiod start date, postperiod start date, preperiod 10-K filing, and postperiod 10-K filing (see Figure 1).	(2,447)
6. Firms without CRSP *PERMNO.*	(954)
7. Firms without valid filings for Perl textual analysis programs as a result of (1) failed extraction of Management Discussion and Analysis and/or financial statement footnotes, (2) file size less than 100 KB, or (3) no HTML in the filing.	(1,131)
8. Firms without available regression variables for all eight firm-quarters in main tests.	(552)
Total number of firms in final sample	2,079

Panel B: Sample distribution by Fama–French 12 (FF12) industry and *DHFIRM*

Firms not affected by SFAS 161 (*DHFIRM* = 0)				Firms affected by SFAS 161 (*DHFIRM* = 1)			
FF12	Industry	*N*	%	FF12	Industry	*N*	%
1	Consumer nondurables	27	3.6	1	Consumer nondurables	56	4.2
2	Consumer durables	11	1.5	2	Consumer durables	28	2.1
3	Manufacturing	40	5.3	3	Manufacturing	136	10.2
4	Energy	17	2.3	4	Energy	79	5.9
5	Chemicals	10	1.3	5	Chemicals	40	3.0
6	Business equipment	161	21.4	6	Business equipment	212	15.9
7	Telecommunications	16	2.1	7	Telecommunications	38	2.9
8	Utilities	11	1.5	8	Utilities	54	4.1
9	Shops	101	13.4	9	Shops	98	7.4
10	Healthcare	117	15.6	10	Healthcare	110	8.3
11	Finance	126	16.8	11	Finance	350	26.4
12	Other	114	15.2	12	Other	127	9.6
	Total	751	100		Total	1,328	100

among investors. I first discuss the main results from estimating Equation (1) with *DHDISC* and *DHACTIVITY* as treatment measures (Table 4). I then consider the robustness of these main results through additional analyses shown in Tables 5–8.

4.3.1. The Effect of SFAS 161 on Information Asymmetry. The first two columns of Table 4 present results with *DHDISC* as the treatment variable, and the last two columns present results using *DHACTIVITY*. For each treatment measure, I present results using the full sample (columns (1) and (3)) and the subsample of DH firms (columns (2) and (4)). All equations are estimated via ordinary least squares with heteroskedasticity-robust standard errors clustered by firm. In Table 4, the coefficients of interest are the difference-in-differences estimates, *POST* × *DHDISC* and *POST* × *DHACTIVITY*, which capture the effect of SFAS 161 on *SPREAD*. Before discussing these coefficients, it is important to note that *DRISK* is significantly positive in all models ($p < 0.01$), suggesting that bid-ask spreads tend to be larger when there are bigger fluctuations in firms' underlying market risk factors. This result lends support to my maintained assumption (discussed in Section 3.1.1) that bid-ask spreads can reflect information asymmetry about firms' risk exposures and DH activities. Also, the adjusted R^2 values are very high for all models (above 0.94), suggesting the *SPREAD* models in Table 4 exhibit excellent fit.

Turning to the coefficients of interest, I first discuss the full-sample results, which include both DH and non-DH firms. Both *POST* × *DHDISC* (column (1)) and *POST* × *DHACTIVITY* (column (3)) are significantly negative ($p < 0.01$), indicating that firms whose disclosures were more likely impacted by SFAS 161 experienced a post-SFAS 161 decrease in information asymmetry (as captured by *SPREAD*). In terms of economic magnitudes, firms with a one-standard-deviation increase in *DHDISC* (*DHACTIVITY*) exhibit an approximately 7% (6%) decrease in *SPREAD* relative to firms with no overall change in their DH footnote disclosures (no derivative activities captured by *DHACTIVITY*).[29] Given the different strengths and

Table 2. Treatment Measures: SFAS 161 DH Disclosure Changes and Pre-SFAS 161 DH Activity

Variable	*N*	Mean	Median	Std. dev.
Non-DH firms (*DHFIRM* = 0)				
Δ*WORDS*	751	−0.007	−0.004	0.099
Δ*NUMS*	751	0.003	0.000	0.057
Δ*GROUP*	751	−0.027	−0.006	0.140
Δ*TABNUMS*	751	1.221	0.000	13.652
DHDISC	751	−0.428	−0.425	0.726
DHACTIVITY	751	0.005	0.000	0.028
DH firms (*DHFIRM* = 1)				
Δ*WORDS*	1,328	0.068	0.024	0.179
Δ*NUMS*	1,328	0.063	0.036	0.115
Δ*GROUP*	1,328	−0.002	0.002	0.056
Δ*TABNUMS*	1,328	11.792	7.185	20.460
DHDISC	1,328	0.242	0.061	1.052
DHACTIVITY	1,328	0.052	0.001	0.121

	t-test for difference in means	Wilcoxon rank-sum test for difference in medians
Δ*WORDS*	0.075 ***	0.028***
Δ*NUMS*	0.059***	0.036***
Δ*GROUP*	0.026***	0.008***
Δ*TABNUMS*	10.571***	7.185***
DHDISC	0.670***	0.486***
DHACTIVITY	0.048***	0.001***

Notes. This table presents descriptive statistics for the DH footnote disclosure characteristics and pre-SFAS 161 DH activity. Because these measures capture the preadoption level of DH activity and *changes* in disclosure characteristics, only one observation per firm is included. Statistics are presented separately for non-DH firms (*DHFIRM* = 0) and DH firms (*DHFIRM* = 1). All variables except *DHDISC* are winsorized at 1% and 99%, and variable definitions are provided in Appendix B.

*$p < 0.10$; **$p < 0.05$; and ***$p < 0.01$ (based on two-tailed tests).

weaknesses of *DHDISC* and *DHACTIVITY* as treatment measures (previously discussed in Section 3.1.2), it is noteworthy that both measures yield similar results in terms of statistical and economic significance. Although I cannot completely rule out alternative factors, the similarity across both treatment measures helps allay concerns that the findings are driven by something other than the adoption of SFAS 161. The results are consistent with investors generally having a better understanding of how DH activities impact firm value, rather than a select few expert investors benefitting from the expanded disclosures.

As discussed in Section 3.1.2, one advantage of using continuous treatment measures such as *DHDISC* and *DHACTIVITY* is that I can employ the same difference-in-differences framework to consider the impact of SFAS 161 only among treated firms (i.e., DH firms). This approach helps ensure that the full-sample results are not driven by confounding factors that differ between DH and non-DH firms. Columns (2) and (4) of Table 4 present results of estimating Equation (1) using only DH firms. It is important that the signs, magnitudes, and statistical significance for *POST* × *DHDISC* (column (2)) and *POST* × *DHACTIVITY* (column (4)) are similar to the coefficients from the full-sample models, suggesting the results are driven by DH disclosure changes imposed by SFAS 161 and not by confounding differences between DH and non-DH firms. To provide additional evidence on this point, I also reestimate the full-sample models from Table 4 using coarsened exact matching and entropy balancing to account for potentially important differences between DH and non-DH firms. For each firm, I use its four pre-SFAS 161 quarterly observations to compute preadoption averages of *SPREAD* and all firm characteristic controls from Equation (1) (i.e., all controls except for *POST* and *VIX*). These preadoption averages are used with coarsened exact matching (using three bins for coarsening all preadoption averages) and entropy balancing (matching on mean and variance for all preadoption averages) to reweight observations so that DH and non-DH firms are similar on important pre-SFAS 161 dimensions. In all cases, *POST* × *DHDISC* and *POST* × *DHACTIVITY* remain negative and significant ($p < 0.01$, untabulated). These tests provide additional evidence that the main results are not driven by confounding factors that differ between DH and non-DH firms.[30]

4.3.2. Parallel Trends Analyses. Table 5 presents graphical and regression analyses to assess trends in *SPREAD* before and after the adoption of SFAS 161. The time series plots in Panels A and B show the movement in *SPREAD* relative to the quarter of SFAS 161 adoption, broken down by quintile for the two treatment measures used in the main results: *DHDISC* (Panel A) and *DHACTIVITY* (Panel B). Each sample firm is assigned to a quintile for *DHDISC* and *DHACTIVITY*, and I take the average *SPREAD* for each quintile and for each quarter in event time (i.e., $QTR_{\text{Adopt-4}}$ through $QTR_{\text{Adopt+3}}$).[31] Finally, because differences in the level of *SPREAD* across quintiles make it difficult to compare trends visually, I subtract the average *SPREAD* corresponding to $QTR_{\text{Adopt-1}}$ for each quintile so that pre- and postadoption trends can be more easily compared visually across quintiles. Overall, the plots in Panels A and B suggest that pre-SFAS 161 trends in *SPREAD* across quintiles were much more similar than the divergent trends observed after SFAS 161 adoption. The plots also show that the decrease in information asymmetry beginning at SFAS 161 adoption is more pronounced for firms with higher values of *DHDISC* and *DHACTIVITY* (i.e., the plots corresponding to higher quintiles of either measure), consistent with the Table 4 results.

I note some signs that preadoption trends in *SPREAD* were not always identical across quintiles,

Table 3. Descriptive Statistics for Regression Variables

Panel A: Full sample descriptive statistics

Variable	*N*	Mean	Std. dev.	5%	25%	Median	75%	95%
SPREAD	16,632	−5.710	1.477	−7.587	−6.803	−6.114	−4.785	−2.752
DRISK	16,632	1.224	0.418	0.678	0.915	1.137	1.470	2.054
POST	16,632	0.500	0.500	0.000	0.000	0.500	1.000	1.000
ROA	16,632	−0.008	0.062	−0.122	−0.006	0.004	0.016	0.042
LEV	16,632	0.581	0.277	0.142	0.366	0.576	0.801	0.949
BTM	16,632	0.902	0.961	0.079	0.374	0.658	1.093	2.672
SIZE	16,632	6.814	1.862	3.704	5.537	6.781	8.014	10.063
PRICE	16,632	18.925	18.716	1.178	5.365	13.370	26.588	54.966
TURNOVER	16,632	−5.144	1.203	−7.580	−5.782	−4.912	−4.325	−3.505
FOLLOW	16,632	1.689	0.976	0.000	1.099	1.792	2.485	3.091
VIX	16,632	30.769	14.304	17.627	20.829	25.276	38.340	63.787
PASTSTDRET	16,632	0.044	0.029	0.014	0.024	0.036	0.056	0.103
TRADESIZE	16,632	249.582	167.663	142.509	163.577	189.383	254.481	598.872

Panel B: Descriptive statistics by *DHFIRM*

	DHFIRM = 0 (*N* = 6,008)			*DHFIRM* = 1 (*N* = 10,624)				
Variable	Mean	Median	Std. dev.	Mean	Median	Std. dev.	Diff. in means	Diff. in medians
SPREAD	−5.403	−5.764	1.472	−5.884	−6.272	1.451	−0.480***	−0.508***
DRISK	1.231	1.144	0.408	1.221	1.134	0.424	−0.010	−0.010**
POST	0.500	0.500	0.500	0.500	0.500	0.500	0.000	0.000
ROA	−0.014	0.004	0.070	−0.005	0.004	0.056	0.009***	0.000***
LEV	0.498	0.474	0.283	0.628	0.627	0.262	0.130***	0.153***
BTM	0.815	0.612	0.840	0.951	0.684	1.020	0.135***	0.072***
SIZE	5.907	5.813	1.720	7.327	7.294	1.740	1.420***	1.481***
PRICE	15.218	10.160	15.844	21.021	15.240	19.856	5.803***	5.080***
TURNOVER	−5.328	−5.120	1.222	−5.040	−4.817	1.180	0.288***	0.303***
FOLLOW	1.479	1.609	0.980	1.807	1.946	0.954	0.329***	0.337***
VIX	30.731	25.334	14.167	30.790	25.276	14.381	0.060	−0.058
PASTSTDRET	0.045	0.037	0.028	0.044	0.036	0.029	−0.001	−0.001***
TRADESIZE	268.293	199.104	183.141	239.001	185.461	157.268	−29.292***	−13.643***

Notes. This table presents descriptive statistics for the regression variables used to estimate Equation (1). All continuous variables are winsorized at 1% and 99%. Detailed variable definitions are provided in Appendix B.
**$p < 0.05$ and ***$p < 0.01$ (based on two-tailed tests).

particularly from $QTR_{\text{Adopt-4}}$ to $QTR_{\text{Adopt-3}}$. Although it is reassuring that the trends appear quite parallel in the quarters immediately preceding SFAS 161 adoption, I further examine this issue with regression analyses. These tests are an important complement to the time series plots in Panels A and B of Table 5 for several reasons. First, the adjusted R^2 values are very high (consistently above 0.94) for all the *SPREAD* models in Table 4, and all the control variables (which tend to differ between DH and non-DH firms—see Table 3, Panel B) load very strongly, even in the presence of firm and time fixed effects. The strong loadings suggest that these controls are important determinants of *SPREAD*, and the time series plots do not account for these associations. Relatedly, because SFAS 161 became effective during the 2008 financial crisis, the need for firm and time fixed effects when assessing preadoption trends is heightened.

For the parallel trends regressions, I include additional time period variables in the models from Table 4 to capture potential movement in *SPREAD* before adoption of SFAS 161. *PRE2* (*PRE1*) is an indicator equal to 1 for observations coming two quarters (one quarter) prior to SFAS 161 adoption. Both variables are added to the models as main effects, and I also include *PRE2* × *DHDISC* and *PRE1* × *DHDISC* in columns (1) and (2) and *PRE2* × *DHACTIVITY* and *PRE1* × *DHACTIVITY* in columns (3) and (4). This framework uses the third and fourth quarters prior to adoption as the basis for comparison, and these are the quarters showing signs of potentially nonparallel trends in Panels A and B of Table 5. If the main results observed in Table 4 are indeed due to adoption of SFAS 161, the additional *PRE2* and *PRE1* interaction terms should not be statistically significant; however, the *POST* × *DHACTIVITY* and *POST* × *DHDISC* coefficients should remain significantly negative. For all four models shown in Panel C of Table 5, none of the additional *PRE2* or *PRE1* interaction terms is significantly different from 0, whereas the *POST* × *DHDISC* and *POST* × *DHACTIVITY* coefficients remain negative and significant ($p < 0.01$).

Table 4. Effect of SFAS 161 on Bid-Ask Spreads Following 10-K and10-Q Filings

	SPREAD			
Sample:	Full	*DHFIRM* = 1	Full	*DHFIRM* = 1
Column:	(1)	(2)	(3)	(4)
POST × *DHDISC*	−0.069***	−0.067***		
	(−7.70)	(−6.28)		
POST × *DHACTIVITY*			−0.057***	−0.046***
			(−5.93)	(−4.66)
POST	0.059**	0.063*	0.082***	0.079**
	(2.13)	(1.65)	(2.95)	(2.04)
DRISK	0.154***	0.147***	0.156***	0.150***
	(6.97)	(5.36)	(7.04)	(5.45)
ROA	−0.386***	−0.364***	−0.382***	−0.354***
	(−4.74)	(−3.43)	(−4.69)	(−3.34)
LEV	0.580***	0.608***	0.572***	0.607***
	(8.65)	(6.65)	(8.56)	(6.60)
BTM	0.140***	0.141***	0.140***	0.143***
	(16.19)	(14.17)	(16.14)	(14.14)
SIZE	−0.259***	−0.255***	−0.267***	−0.262***
	(−9.39)	(−7.89)	(−9.59)	(−8.10)
PRICE	−0.007***	−0.007***	−0.008***	−0.007***
	(−8.01)	(−7.81)	(−8.19)	(−8.15)
TURNOVER	−0.217***	−0.201***	−0.215***	−0.198***
	(−22.82)	(−16.35)	(−22.58)	(−16.03)
FOLLOW	−0.126***	−0.140***	−0.128***	−0.145***
	(−7.30)	(−6.27)	(−7.42)	(−6.45)
VIX	0.022***	0.024***	0.022***	0.024***
	(17.91)	(14.35)	(17.79)	(14.29)
PASTSTDRET	3.312***	3.532***	3.278***	3.496***
	(14.68)	(12.03)	(14.48)	(11.83)
TRADESIZE	0.001***	0.001***	0.001***	0.001***
	(14.32)	(11.10)	(14.11)	(10.87)
Firm and time FE	Yes	Yes	Yes	Yes
N	16,632	10,624	16,632	10,624
Adj. R^2	0.944	0.942	0.944	0.942

Notes. This table presents estimations of Equation (1) to document the effect of SFAS 161 adoption on *SPREAD*, the natural log of the average daily bid-ask spread during the calendar month following the 10-Q or 10-K filing date. Column headings indicate the sample used for each model. Variable definitions are provided in Appendix B. All continuous variables except *DHDISC* are winsorized at 1% and 99%. Standard errors are heteroskedasticity robust and clustered by firm; *t*-statistics are reported in parentheses. FE, fixed effects.

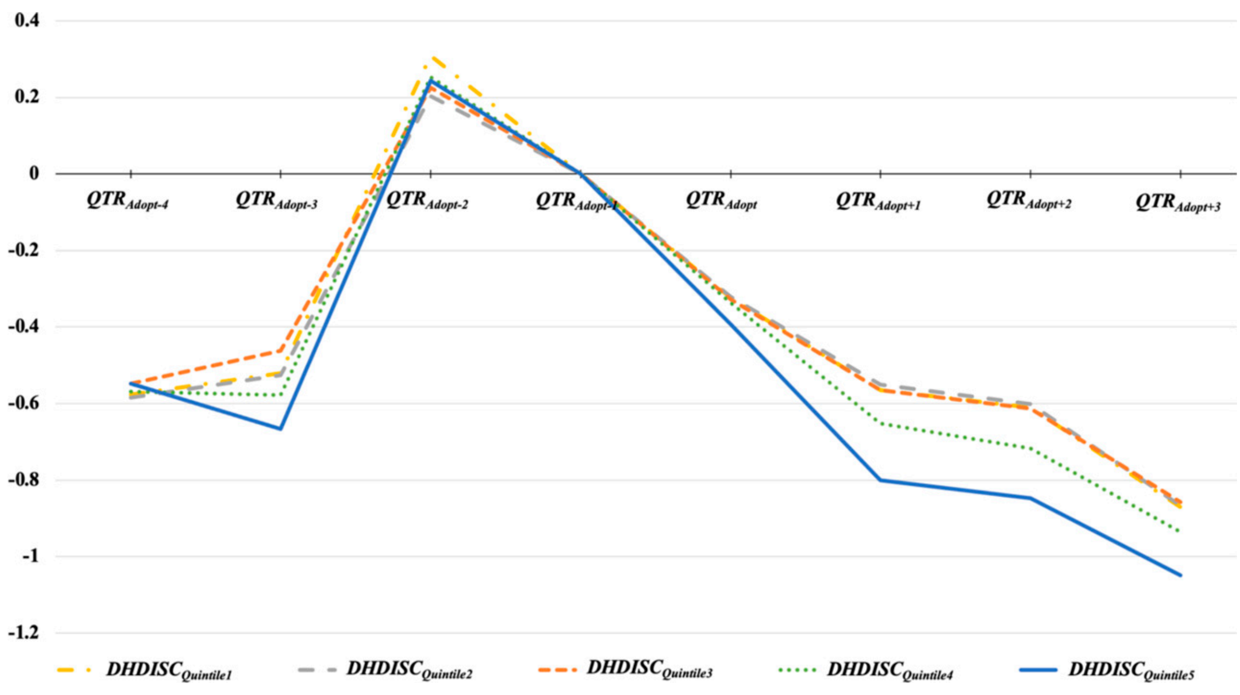
$^{*}p < 0.10$; $^{**}p < 0.05$; and $^{***}p < 0.01$ (based on two-tailed tests).

Ultimately, the results in Table 5 cannot verify the (untestable but maintained) assumption that *SPREAD* would have continued on the same trend for less affected and more affected firms in the absence of SFAS 161. However, the Table 5 analyses help bolster inferences stemming from the main results by suggesting that, overall, firms whose DH disclosures were more likely to have changed by adopting SFAS 161 did not exhibit significant differential preadoption movements in *SPREAD* relative to firms whose DH disclosures were less likely to have changed after adopting SFAS 161.

4.3.3. Placebo Adoption Analyses. Next, I perform two complementary placebo adoption tests to provide additional evidence that the main results in Table 4 are driven by firms adopting SFAS 161. Results in Table 6, Panel A, are based on an entirely new sample centered on a pseudo-effective date of November 15, 2011 (rather than SFAS 161's true effective date of November 15, 2008). I use a future date rather than a past date to avoid the period when investors were expressing concern about DH disclosures, because firms could have been differentially responding to investors' demands for information (e.g., the National City Corporation example mentioned in Section 3.1.2). I chose the pseudo-effective date to ensure that the placebo analyses did not overlap with my main sample period, and I form a new sample using the same procedures as the main sample (see Figure 1) but centered on the pseudo-effective date. I recalculate all regression variables, including the *DHFIRM* and *DHACTIVITY* treatment measures. If the main results in Table 4 are indeed attributable to the adoption of SFAS 161, I should find no significant results for *POST* × *DHDISC* or *POST* × *DHACTIVITY*. Panel A of Table 6 confirms this expectation with none of these coefficients being significantly different from 0.

Table 5. (Color online) Parallel Trends Analyses

Panel A: Quarterly trends in *SPREAD* by *DHDISC* quintile



Panel B: Quarterly trends in *SPREAD* by *DHACTIVITY* quintile

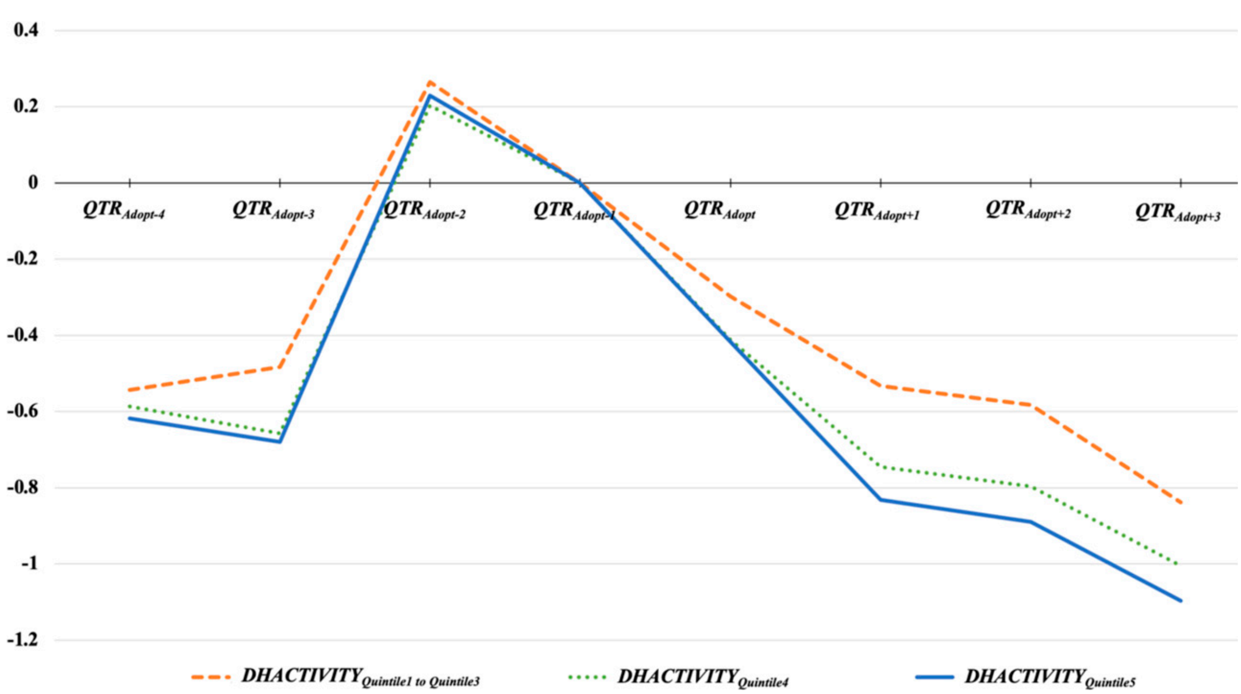


Table 5. (Continued)

Panel C: Parallel trends regression analyses				
	SPREAD			
Sample:	Full	*DHFIRM* = 1	Full	*DHFIRM* = 1
Column:	(1)	(2)	(3)	(4)
PRE2 × *DHDISC*	−0.003	−0.010		
	(−0.31)	(−0.76)		
PRE1 × *DHDISC*	0.002	−0.009		
	(0.14)	(−0.67)		
POST × *DHDISC*	−0.069***	−0.071***		
	(−6.86)	(−6.03)		
PRE2 × *DHACTIVITY*			0.009	−0.001
			(0.84)	(−0.06)
PRE1 × *DHACTIVITY*			0.006	−0.003
			(0.47)	(−0.20)
POST × *DHACTIVITY*			−0.053***	−0.047***
			(−5.12)	(−4.41)
Controls	Yes	Yes	Yes	Yes
Firm and time FE	Yes	Yes	Yes	Yes
N	16,632	10,624	16,632	10,624
Adj. R^2	0.944	0.942	0.944	0.942

Notes. The figures in Panels A and B present trends in *SPREAD* (the outcome variable in Table 4) relative to the quarter of SFAS 161 adoption, broken down by quintile for the two treatment measures used in the main results: *DHDISC* (Panel A) and *DHACTIVITY* (Panel B). Each sample firm is assigned to a quintile for both *DHDISC* and *DHACTIVITY*, and I then take the mean *SPREAD* for each quintile for each quarter in event time (i.e., $QTR_{Adopt-4}$ through $QTR_{Adopt+3}$). Because *DHACTIVITY* = 0 for 63% of the sample, I group these observations together in Panel B and label them $DHACTIVITY_{Quintile1\ to\ Quintile3}$, and the remaining observations with non-zero *DHACTIVITY* are divided into two groups with each having roughly the same number of observations as each *DHDISC* quintile. Finally, for each quintile, I subtract the average *SPREAD* corresponding to *QTRAdopt-1* so that pre- and post-adoption trends can be more easily compared visually across quintiles. Panel C table repeats the specifications from Table 4 after including additional time period variables. *PRE2* (*PRE1*) is an indicator variable equal to 1 for observations two quarters (one quarter) prior to SFAS 161 adoption. *PRE2* and *PRE1* are included as main effects and as interactions with both *DHDISC* and *DHACTIVITY*. Standard errors are heteroskedasticity robust and clustered by firm; *t*-statistics are reported in parentheses. FE, fixed effects.
*$p < 0.10$; **$p < 0.05$; and ***$p < 0.01$ (based on two-tailed tests).

The results in Table 6, Panel A, provide additional support for attributing the main findings in Table 4 to the adoption of SFAS 161, but this out-of-sample approach only demonstrates that there is no detectable effect for a sample centered on one particular pseudo-effective date (i.e., November 15, 2011). Although it is not feasible to construct entirely new samples for many pseudo-effective dates, I employ a complementary bootstrap approach using the main sample from Table 4 to provide additional evidence that my results are driven by SFAS 161 adoption. I perform 10,000 repetitions of the models in Table 4 after randomly assigning each firm's quarter of SFAS 161 adoption (i.e., one of each firm's eight sample quarters is chosen randomly for each bootstrap repetition—see Figure 1). Using the randomly chosen adoption date, *POST* is redefined for each firm and each repetition. For each of the 10,000 repetitions, I retain the estimates of *POST* × *DHDISC* and *POST* × *DHACTIVITY* in order to construct distributions of the bootstrap coefficients. If SFAS 161 is driving my results, then the Table 4 coefficients (which are negative) should be in the left tails of the distributions of the 10,000 bootstrap coefficients. Moreover, the means of the 10,000 bootstrap coefficients should be closer to 0 than the corresponding coefficients in Table 4. Stated differently, if the main results are *not* attributable to SFAS 161 adoption but are instead driven by some other factor, then firms' actual adoption dates should not be important for reproducing the main results using the same firm-quarters as the main analyses. Under this alternative scenario, the means of the 10,000 bootstrap coefficients would not be statistically different from the corresponding coefficients presented in Table 4.

Panel B of Table 6 presents the bootstrap results, which support the conclusion that SFAS 161 adoption matters for the main results shown in Table 4. Specifically, the means of the 10,000 bootstrap coefficients are all closer to 0 relative to their corresponding coefficients from Table 4. Furthermore, the differences between the means of the 10,000 bootstrap coefficients and the coefficients from Table 4 are highly significant at $p < 0.01$ for all specifications (i.e., both *DHDISC* and *DHACTIVITY* models for both the full sample and *DHFIRM* = 1 subsample). Finally, the Table 4 coefficients land in the left tail for each of their respective bootstrap coefficient distributions: depending on the model considered, 96.9%, 94.2%, 94.2%, or 87.6% of the 10,000 bootstrap coefficients are closer to 0 than the coefficients from Table 4.[32] Coupled with the out-

Table 6. Placebo Analyses

Panel A: Out-of-sample placebo effective date for SFAS 161

	SPREAD			
Sample:	Full	*DHFIRM* = 1	Full	*DHFIRM* = 1
Column:	(1)	(2)	(3)	(4)
POST × *DHDISC*	−0.003	−0.008		
	(−0.59)	(−1.33)		
POST × *DHACTIVITY*			0.001	0.005
			(0.12)	(0.79)
Controls	Yes	Yes	Yes	Yes
Firm and time FE	Yes	Yes	Yes	Yes
N	21,784	15,184	21,728	15,152
Adj. R^2	0.973	0.974	0.973	0.974

Panel B: Bootstrap results using random firm-specific adoption dates within the main sample period

Coefficient of interest	Sample	Coefficient estimate from Table 4	Mean of 10,000 bootstrap coefficients	Std. dev. of 10,000 bootstrap coefficients	Difference between mean of bootstrap coefficients and Table 4 coefficient	% of 10,000 bootstrap coefficients closer to 0 than Table 4 coefficient
POST × *DHDISC*	Full	−0.069	−0.058	0.006	0.011***	96.9
POST × *DHDISC*	*DHFIRM* = 1	−0.067	−0.056	0.007	0.011***	94.2
POST × *DHACTIVITY*	Full	−0.057	−0.047	0.006	0.010***	94.2
POST × *DHACTIVITY*	*DHFIRM* = 1	−0.046	−0.039	0.006	0.007***	87.6

Notes. Panel A presents regression results after recalculating all variables for a sample of firm-quarters centered on a placebo effective date of November 15, 2011 (instead of the actual SFAS 161 effective date of November 15, 2008). Panel B compares the coefficients of interest from Table 4 with the average of bootstrap coefficients generated from 10,000 repetitions of the models from Table 4 with randomly assigned firm-specific adoption dates within the main sample period. Standard errors are heteroskedasticity robust and clustered by firm; t-statistics are reported in parentheses. FE, fixed effects.

***$p < 0.01$ (based on two-tailed tests).

of-sample analyses in Panel A, the bootstrap results in Table 6, Panel B, provide additional evidence that the adoption of SFAS 161 led to reduced information asymmetry among investors.

4.3.4. Ruling Out Potentially Confounding Real Effects. To address the concern that changes in firms' DH behavior caused by the 2008 financial crisis and/or SFAS 161 adoption (i.e., real effects) are potentially driving my results and confounding my inferences, I first investigate whether firms' overall risk exposure to market risk factors (e.g., interest rates, foreign exchange rates, or commodity prices) changes after the adoption of SFAS 161. As explained in more detail in Appendix B and Section IA.3 of the internet appendix, I generate firm-specific exposure weights for 17 risk factors related to interest rates, foreign exchange rates, energy prices, and nonenergy commodity prices (the W_{ir} in Equation (IA.1) in the internet appendix). If firms significantly change their DH and/or risk management activities after SFAS 161, their calculated risk exposure weights should reflect the change(s). To test whether such changes are driving my results, I calculate each firm's 17 W_{ir} weights using data from 2007 to 2008 (the two calendar years before SFAS 161 became effective) and again from 2009 to 2010 (the two calendar years after SFAS 161 became effective). $RISKEXP_{\mathrm{PRE}}$ ($RISKEXP_{\mathrm{POST}}$) is the sum of each firm's 17 W_{ir} exposure weights measured from 2007 to 2008 (2009 to 2010). Because each of the 17 weights can take the value of 0, 1, or 2, the maximum value of $RISKEXP_{\mathrm{PRE}}$ and $RISKEXP_{\mathrm{POST}}$ is 34. I then define $\Delta RISKEXP$ ($|\Delta RISKEXP|$) as the (absolute) difference between $RISKEXP_{\mathrm{POST}}$ and $RISKEXP_{\mathrm{PRE}}$.

Table 7 provides descriptive statistics for these risk exposure measures separately for non-DH firms (*DHFIRM* = 0) and DH firms (*DHFIRM* = 1) along with tests for differences in means in medians. The results in Table 7 show that in both the pre- and post-SFAS 161 periods, DH firms have significantly lower overall risk exposure ($p < 0.01$ for tests of differences in means and medians). This finding is consistent with derivatives being used primarily to reduce risk and not to speculate. More important, I find no significant difference between DH and non-DH firms for the mean or median of $\Delta RISKEXP$ or $|\Delta RISKEXP|$. This finding indicates that, on average, DH firms' risk exposures did not *change* differently than non-DH firms' risk exposures around the adoption of SFAS 161. Put differently, if DH firms were materially changing their DH activities as a result of the financial crisis and/or SFAS 161, we should observe significantly different values of $\Delta RISKEXP$ and/or $|\Delta RISKEXP|$ (indicating shifts in risk exposure) for DH firms relative to non-

Table 7. Total Risk Exposure Before and After SFAS 161

Variable	*N*	Mean	Median	Std. dev.
Non-DH firms (*DHFIRM* = 0)				
$RISKEXP_{PRE}$	747	17.647	18.000	7.337
$RISKEXP_{POST}$	747	17.695	18.000	7.274
$\Delta RISKEXP$	747	0.048	0.000	7.691
$\lvert\Delta RISKEXP\rvert$	747	6.203	5.000	4.541
DH firms (*DHFIRM* = 1)				
$RISKEXP_{PRE}$	1,324	16.622	17.000	7.595
$RISKEXP_{POST}$	1,324	16.619	17.000	7.665
$\Delta RISKEXP$	1,324	−0.004	0.000	7.780
$\lvert\Delta RISKEXP\rvert$	1,324	6.128	5.000	4.791

	t-test for difference in means	Wilcoxon rank-sum test for difference in medians
$RISKEXP_{PRE}$	−1.024***	−1.000***
$RISKEXP_{POST}$	−1.076***	−1.000***
$\Delta RISKEXP$	−0.052	0.000
$\lvert\Delta RISKEXP\rvert$	−0.076	0.000

Notes. This table presents a comparison of overall risk exposure between non-DH firms (*DHFIRM* = 0) and DH firms (*DHFIRM* = 1) before and after SFAS 161. $RISKEXP_{PRE}$ ($RISKEXP_{POST}$) is a measure of each firm's overall risk exposure before (after) SFAS 161. These measures are constructed in the same way as the denominator of equation (IA.1) in the internet appendix except that instead of using data from 2005 to 2007, I use data from 2007 to 2008 for $RISKEXP_{PRE}$ and 2009 to 2010 for $RISKEXP_{POST}$. In each period, I run 17 regressions for each firm in order to estimate its exposure to each of 17 different risk factors related to interest rates, foreign exchange rates, energy prices, and nonenergy commodity prices. I use the absolute coefficient estimates on each risk factor to divide the firms into separate terciles for each of the 17 risk factors; this means a firm could be in the bottom tercile for one factor and the top or middle tercile for another. Firms in the bottom tercile for a particular factor are assigned a weight of 0 for that factor, representing little to no exposure. Similarly, firms in the middle (top) tercile for a particular factor are assigned a weight of 1 (2) for that factor, capturing medium (high) exposure. For each firm, $RISKEXP_{PRE}$ ($RISKEXP_{POST}$) is the sum of the firm's 17 weights based on data from 2007 to 2008 (2009 to 2010) and represents the firm's overall level of risk exposure in the respective period. $\Delta RISKEXP$ is the difference between $RISKEXP_{POST}$ and $RISKEXP_{PRE}$, and $\lvert\Delta RISKEXP\rvert$ is the absolute value of $\Delta RISKEXP$.

***$p < 0.01$ (based on two-tailed tests).

DH firms. The evidence that these values do *not* significantly differ helps allay concerns that material changes in firms' risk management and/or DH activities are driving the main results in Table 4.

Notwithstanding the evidence in Table 7, I also reestimate the models from Table 4 after directly controlling for *changes in* several firm characteristics from pre- to post-SFAS 161 adoption. Because the Table 4 results include firm fixed effects, any confounding factors potentially related to firms' DH activities and/or risk management would need to *change* upon adopting SFAS 161. I consider several factors (see Appendix B for detailed definitions): Δ*RISKEXP* (the change in overall risk exposure analyzed in Table 7), Δ*10KSIZE* (the change in the 10-K size), Δ*LINECREDIT* (the change in the importance of lines of credit), Δ*CRISIS* (the change in the impact of the 2008 financial crisis), Δ*NOTESPAY* (the change in the use of short-term notes payable), Δ*CRATING* (the change in the investment-grade status of a firm's credit rating), and Δ*CASH* (the change in the amount of cash on hand). Note that Δ*RISKEXP* is motivated by the idea that material changes in DH programs would manifest in firms' risk exposures and by Chiorean's (2016) evidence that firms decrease (particularly speculative) derivative use after SFAS 161; Δ*10KSIZE* is motivated by the concern that *DHDISC* is simply capturing overall increases in disclosure unrelated to SFAS 161. The other variables are motivated by Calluzzo and Dudley's (2018) evidence that firms without investment-grade credit ratings were "more likely to lose access to over-the-counter derivatives" (p. 1) when their lenders were affected by the 2008 financial crisis and that firms that lost access relied more on lines of credit and cash savings for liquidity.

Table 8 presents results of analyses that include these additional controls.[33] Panel A presents Pearson and Spearman correlations for each of the real effects variables with *DHDISC* and *DHACTIVITY*. Δ*RISKEXP* is significantly negatively correlated with *DHDISC* and *DHACTIVITY*, suggesting that firms more likely to have overall DH disclosure changes from SFAS 161 also tend to have reduced overall risk exposures after SFAS 161; Δ*CRISIS* and Δ*NOTESPAY* also exhibit some negative correlations, but none of the other real effects variables is significantly correlated with the two treatment measures. Similar to Table 4, Panel B of Table 8 presents four models (for both *DHDSIC* and *DHACTIVITY* using both the full and DH firm samples) after including each real effects control interacted with *POST*. If real effects are responsible for my main results, I would expect my two coefficients of interest (*POST* × *DHDISC* and *POST* × *DHACTIVITY*) to either stop loading or move materially closer to 0 relative to the Table 4 coefficients. Thus, it is important that both *POST* × *DHDISC* and

Table 8. Effect of SFAS 161 on Bid-Ask Spreads After Controlling for Potential Real Effects

Panel A: Correlations of *DHDISC* and *DHACTIVITY* with real effects variables (N = 2,071 firms)

	Δ*RISKEXP*	Δ*10KSIZE*	Δ*LINECREDIT*	Δ*CRISIS*	Δ*NOTESPAY*	Δ*CRATING*	Δ*CASH*
Pearson for *DHDISC*	−0.075**	−0.026	0.009	−0.047**	−0.011	−0.023	−0.023
Spearman for *DHDISC*	−0.064**	−0.037	−0.010	−0.038	−0.053**	−0.019	−0.039
Pearson for *DHACTIVITY*	−0.072**	−0.019	−0.007	−0.094**	−0.018	0.027	−0.017
Spearman for *DHACTIVITY*	−0.038	−0.043	−0.024	−0.098**	−0.099**	0.032	−0.008

Panel B: Replication of main results including real effects control variables

	SPREAD			
Sample:	Full	*DHFIRM* = 1	Full	*DHFIRM* = 1
Column:	(1)	(2)	(3)	(4)
POST × *DHDISC*	−0.068***	−0.065***		
	(−7.60)	(−6.04)		
POST × *DHACTIVITY*			−0.055***	−0.044***
			(−5.76)	(−4.42)
POST × Δ*RISKEXP*	0.003**	0.003**	0.003**	0.004***
	(2.42)	(2.36)	(2.56)	(2.59)
POST × Δ*10KSIZE*	−0.000	−0.000	−0.000	−0.000
	(−0.77)	(−0.27)	(−0.67)	(−0.20)
POST × Δ*LINECREDIT*	0.324*	0.421*	0.305	0.396
	(1.69)	(1.77)	(1.55)	(1.61)
POST × Δ*CRISIS*	0.611	0.226	0.560	0.235
	(1.46)	(0.44)	(1.30)	(0.45)
POST × Δ*NOTESPAY*	0.010***	0.008**	0.010***	0.008**
	(3.19)	(2.38)	(3.10)	(2.24)
POST × Δ*CRATING*	0.021	0.049	0.040	0.072
	(0.41)	(0.89)	(0.79)	(1.35)
POST × Δ*CASH*	−0.002	0.000	−0.001	0.000
	(−1.07)	(0.23)	(−1.01)	(0.24)
Controls	Yes	Yes	Yes	Yes
Firm and time FE	Yes	Yes	Yes	Yes
N	16,568	10,592	16,568	10,592
Adj. R^2	0.945	0.943	0.944	0.942

Notes. This table presents analyses to rule out potentially confounding real effects as drivers of the main results. I employ several controls for real effects, and each is measured as the *firm-level change* from the pre- to post-SFAS 161 period. Section 4.3.4 discusses the real effects variables, and Appendix B provides detailed definitions. Panel A presents Pearson and Spearman correlations between the real effects variables and *DHDISC* and *DHACTIVITY*. Panel B repeats regression models from Table 4 after including the real effects controls interacted with *POST*. All control variables shown in Table 4 are included in the models presented here in Panel B but are not shown for parsimony. Standard errors are heteroskedasticity robust and clustered by firm; t-statistics are reported in parentheses. FE, fixed effects.

**$p < 0.05$ in Panel A; *$p < 0.10$; **$p < 0.05$; and ***$p < 0.01$ (based on two-tailed tests) in Panel B.

POST × *DHACTIVITY* remain significantly negative ($p < 0.01$), and the coefficient magnitudes scarcely change when compared with Table 4.[34] Altogether, the results in Tables 7 and 8 suggest that the main results are not driven by confounding real effects, lending additional support for the conclusion that SFAS 161 resulted in less information asymmetry among investors.

5. Additional Analyses: Disclosure Content vs. Disclosure Format

The results in Tables 4–8 suggest that SFAS 161 disclosures led to lower information asymmetry among investors (as captured by bid-ask spreads). Table 9 presents results of estimating Equation (1) with Δ*WORDS*, Δ*NUMS*, Δ*GROUP*, and Δ*TABNUMS* as alternative *TREAT* measures to address my secondary research question of which footnote disclosure characteristics matter most for the decreased information asymmetry documented in Tables 4–8.[35] Throughout Table 9, *POST* × Δ*WORDS*, *POST* × Δ*NUMS*, and *POST* × Δ*TABNUMS* are significantly negative ($p < 0.01$) when included separately, suggesting that increased DH disclosure content and more tabular display each contribute to the reduction in information asymmetry shown in Tables 4–8. The *POST* × Δ*GROUP* coefficient is never significant, consistent with disclosure grouping mattering less for information asymmetry. Table 9 also presents specifications including all four disclosure change variables simultaneously, and only *POST* × Δ*WORDS* and *POST* × Δ*NUMS* continue to load negatively ($p < 0.01$). This evidence suggests that disclosure content mattered more than disclosure format for reducing information asymmetry after SFAS 161. Moreover, the coefficient

Table 9. Analysis of Footnote Disclosure Content vs. Format

	SPREAD									
Sample:	Full					*DHFIRM* = 1				
Column:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
POST × Δ*WORDS*	−0.072***				−0.046***	−0.061***				−0.035***
	(−7.76)				(−3.62)	(−5.93)				(−2.60)
POST × Δ*NUMS*		−0.073***			−0.044***		−0.063***			−0.039***
		(−7.54)			(−3.41)		(−5.90)			(−2.89)
POST × Δ*GROUP*			−0.003		0.008			−0.017		0.001
			(−0.68)		(1.55)			(−1.53)		(0.11)
POST × Δ*TABNUMS*				−0.039***	−0.009				−0.033***	−0.009
				(−3.92)	(−0.85)				(−2.80)	(−0.69)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm and time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	16,632	16,632	16,632	16,632	16,632	10,624	10,624	10,624	10,624	10,624
Adj. R^2	0.944	0.944	0.944	0.944	0.944	0.942	0.942	0.942	0.942	0.942

Notes. This table presents the results of reestimating the *SPREAD* models from Table 4 using changes in specific DH footnote disclosure characteristics as alternative *TREAT* variables. All control variables shown in Table 4 are included in the models presented here, but they are not shown for parsimony. Standard errors are heteroskedasticity robust and clustered by firm; *t*-statistics are reported in parentheses. FE, fixed effects.
***$p < 0.01$ (based on two-tailed tests).

point estimates for *POST* × Δ*WORDS* and *POST* × Δ*NUMS* are very close to each other (untabulated Wald tests confirm that these coefficients are not significantly different from each other in models including all variables simultaneously). This evidence suggests that more qualitative disclosure and more disaggregated quantitative content similarly contribute to the decreased information asymmetry observed for firms more strongly impacted by the adoption of SFAS 161.

6. Conclusion

I analyze the information asymmetry effects of SFAS 161's expanded DH footnote disclosures. Using textual analysis to measure changes in the content and format of firms' actual DH footnote disclosures, I show that firms whose DH disclosures changed more after adopting SFAS 161 exhibited a greater decrease in bid-ask spreads. Using a complementary approach to capture the firm-specific impact of SFAS 161 that is based on firms' pre-SFAS 161 levels of DH activities, I find similar results. Taken together, this evidence is consistent with SFAS 161 disclosures leading to less information asymmetry among investors. This inference is supported by robustness tests related to parallel trends in bid-ask spreads before SFAS 161 adoption, pseudo-adoption dates, and potentially confounding real effects. My results also point to the qualitative and quantitative content of DH disclosures having greater impact on information asymmetry than DH disclosure format. Although my overall results suggest that SFAS 161 disclosures resulted in less information asymmetry among investors, this conclusion is subject to the important caveat that I cannot completely rule out the possibility that some unidentified factor besides SFAS 161 is responsible for the documented decrease in bid-ask spreads.

Although the FASB's stated objective of SFAS 161 was not to impact bid-ask spreads or information asymmetry per se, I believe my findings speak to the Board's desired outcome of improving investors' understanding. If SFAS 161 led to more confusion among investors (a possibility that was suggested in several comment letters), or if only a select few expert investors improved their understanding after SFAS 161, I would have expected to find an increase in bid-ask spreads for firms most affected by the new standard. My robust evidence that bid-ask spreads decreased for firms whose disclosures were more likely affected by SFAS 161 is consistent with investors generally improving their understanding of how firms' DH activities impact firm value. Because the FASB has changed required DH disclosure multiple times over the past few decades, it is possible that DH disclosures will be reconsidered in the future, and my evidence about specific aspects of SFAS 161 (i.e., content versus format) may prove useful. Moreover, Lev's (1988) line of reasoning that disclosure regulation can be evaluated through its impact on information asymmetry, coupled with regulators' efforts to "level the playing field" for all investors, suggests that my results may be of interest to standard setters and regulators as they consider disclosure requirements in other contexts (depending on how each context compares to DH activities and the details of SFAS 161).

This paper contributes to the literature by studying the effects of SFAS 161 disclosures for a broad sample of firms across many industries. My approach may prove useful for researchers considering the impact of disclosures about other complex topics typically requiring manual data collection and small samples. In addition, I develop methods using textual analysis and a more comprehensive DH dictionary to shed light on four distinct characteristics of DH disclosures: the amount of qualitative information, disaggregation of quantitative information, grouping of similar information, and tabular display of quantitative information. Future research could employ these methods to study the relations among these (and other) disclosure characteristics. In addition, researchers might consider the extent to which certain characteristics are more or less prevalent in disclosures related to other topics in firms' financial reports to shed additional light on how disclosures impact a firm's information environment.

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Appendix A. Summary of SFAS 133 and SFAS 161 Disclosure Requirements

Disclosure type	SFAS 133 requirements	SFAS 161 requirements
Objectives of derivative and hedging activities	• Classification by hedge type then by risk management policy • Description of purposes of nonhedging derivative activity • Additional qualitative risk management disclosures are encouraged	• Informs preparers that the disclosures' objective is users' understanding of derivative and hedging activities • Classification by underlying risk exposure, then by risk management purposes, then by hedge type • Additional qualitative risk management disclosures are encouraged • Description of purposes of nonhedging derivative activity • Information about the volume of derivative activity
Balance sheet information	• Fair values separated by assets vs. liabilities (SFAS 107) • Clear link to what is reported on the balance sheet (SFAS 107)	• Fair value of derivatives separated by (1) assets vs. liabilities, (2) hedging vs. nonhedging, and (3) risk exposure category • Balance sheet line items that contain the fair values of the various categories
Income statement information	• Fair value hedges: ○ Net gain or loss from hedge ineffectiveness, exclusion from effectiveness testing, and hedge disqualification • Cash flow hedges: ○ Net gain or loss from hedge ineffectiveness and exclusion from effectiveness testing ○ Gross gains and losses from hedge disqualification ○ Description of events that reclassify gains and losses from AOCI to earnings and an estimate of the amount to be reclassified within 12 months ○ Maximum length of time over which non-variable-interest cash flow variability is hedged • Net investment hedges: ○ Net gain or loss from effective hedging recognized in OCI • Description of where these gains and losses are reported on the income statement • Similar disclosures for related financial instruments are encouraged	• Fair value hedges: ○ Net gain or loss from hedge ineffectiveness, exclusion from effectiveness testing, and hedge disqualification ○ Gross gains and losses by risk exposure category from hedging instruments and related hedged items • Cash flow hedges: ○ Gross gains and losses by risk exposure category from effective hedging recognized in OCI, effective hedging reclassified from AOCI, hedge ineffectiveness, exclusion from effectiveness testing, and hedge disqualification ○ Description of events that reclassify gains and losses from AOCI to earnings and an estimate of the amount to be reclassified within 12 months ○ Maximum length of time over which non-variable-interest cash flow variability is hedged • Net investment hedges: ○ Gross gains and losses by risk exposure category from effective hedging recognized in OCI, effective hedging reclassified from AOCI, hedge ineffectiveness, and exclusion from effectiveness testing • Nonhedging derivatives: ○ Gross gains and losses by risk exposure category, or

Appendix A. (Continued)

Disclosure type	SFAS 133 requirements	SFAS 161 requirements
		○ Firms can combine gains and losses from nonhedging derivatives with gains and losses from other trading instruments by risk exposure category • Income statement line items that contain these various gains and losses • Similar disclosures for related financial instruments are encouraged
Credit-risk-related contingent features	• Not required	• Existence and nature of these features and how the features might be triggered in derivatives that are net liabilities • Aggregate fair values of derivatives with such features that are net liabilities • Aggregate fair value of assets that are posted as collateral, would be posted as collateral, or would be required to settle the instrument if such features were triggered
Frequency	• Annual	• Annual and interim
Cross-referencing	• Not required	• Required if information is presented in more than one footnote
Presentation format	• No guidelines	• Tabular format for balance sheet and income statement information

Note. AOCI, accumulated other comprehensive income; OCI, other comprehensive income.

Appendix B. Variable Definitions

Variable	Definition
	Outcome variable
SPREAD	The natural log of the average daily bid-ask spread during the calendar month following the 10-Q or 10-K filing date. For each firm-day, the bid-ask spread is measured from the CRSP Daily Stock File as *ASK-BID* scaled by the average of *ASK* and *BID*.
	Treatment variables
DHDISC	The first (and only) factor with an eigenvalue greater than 1 emerging from a principal component factor analysis of $\Delta WORDS$, $\Delta NUMS$, $\Delta GROUP$, and $\Delta TABNUMS$ (see Figure 1 and related definitions in what follows).
DHACTIVITY	The average value of $100 \times \lvert CIDERGLQ/ATQ \rvert$ using Compustat data over the eight quarters prior to the start of the main sample period (see Figure 1). For each firm-quarter, *CIDERGLQ* is set to 0 if it is missing.
	DH disclosure content and format variables
WORDS	The percentage of all words in the financial statement footnotes matching the DH dictionary shown in Table IA.5 of the internet appendix. Total words are identified by counting all instances of words in the *Loughran–McDonald 2016 Master Dictionary* (available at https://sraf.nd.edu/textual-analysis/resources/#Master%20Dictionary, accessed June 16, 2017) and adding it to the count of all DH dictionary matches. Before counting words and word percentages, the spaces are removed in all dictionary phrases in all filings (i.e., "gas collars" is replaced with "gascollars") so that each instance of each phrase contributes once to both the numerator and denominator when calculating the dictionary percentage for each filing. To construct the DH dictionary, I copied the derivative footnote text from the initial 10-Q filing under SFAS 161 for 94 derivative-user firms and counted the frequency of all unique words, two-word phrases, and three-word phrases appearing in this corpus of 94 derivative footnotes. I manually inspected each word appearing at least 100 times in the entire corpus (230 words corresponding to 76.27% of all words in the corpus). For unambiguous words such as "derivative" and "hedge," the words were added to the dictionary. However, for ambiguous words such as "interest" and "contracts," the lists of unique phrases appearing at least five times in the corpus were manually inspected for any phrase that contained the word of interest. If the phrase pertained to DH activities, the phrase was added to the dictionary but not the original word. For example, "forward interest rate" is in the dictionary, but "interest" is not. For acronyms, I looked at DH-related acronyms in the glossary of the December 2016 *BIS Quarterly Review* (https://www.bis.org/publ/qtrpdf/r_qt1612c.pdf, accessed December 8, 2021) and the list at https://www.derivativepath.com/commonly-used-acronyms (last accessed December 8, 2021). The dictionary items and their frequencies are listed in Table IA.5 of the internet appendix. The final DH dictionary comprises 231 words, phrases, and acronyms.
$\Delta WORDS$	*WORDS* from the postperiod 10-K footnotes minus *WORDS* from the preperiod 10-K footnotes.

Appendix B. (Continued)

Variable	Definition
NUMS	For each filing, I perform two separate procedures. First, I strip the filing of all HTML tags and then use Perl's Lingua::EN::Sentence module to parse the text into sentences. For all sentences with at least three words, the sentence is checked to see whether it matches a dictionary item. If it does, the count of all numbers in the sentence is retained after removing numbers related to dates, accounting standards, footnotes, and Level 1/2/3 financial instruments (numbers are identified with the Perl regular expression "[\s\-\$\(]\d+"). Second, I reanalyze the filing and retain the HTML tags in order to identify tables containing DH information. For each HTML table, I check each cell for purely numeric content (the Perl regular expression is "\A[\$\(\-\s]*[0123456789,\.]+[\)\s]*\z"). If the cell only contains numeric content, I check all cells in the same column directly above the numeric cell and all cells in the same row directly behind (to the left of) the numeric cell. If any of these cells contain dictionary items, the numeric cell is counted as a DH number. *NUMS* is the sum of the counts from these two procedures, multiplied by 100 and scaled by the total word count in the 10-K footnotes. In other words, it is the scaled count of DH numbers, whether they appear in sentences or in tables.
Δ*NUMS*	*NUMS* from the postperiod 10-K footnotes minus *NUMS* from the preperiod 10-K footnotes.
GROUP	The average reduced frequency (ARF) of DH dictionary items occurring in the 10-K footnotes, multiplied by −1 (to maintain consistent signs with the other disclosure change variables). To summarize the approach, if *n* DH dictionary items exist in the document, the document is divided into *n* sections of equal length. Each section is then examined to see whether it contains at least one DH dictionary item. The reduced frequency is the number of sections containing at least one dictionary item, scaled by *n*. Following Allee and DeAngelis (2015), I use the *average* reduced frequency, which accounts for all possible section boundaries. See Allee and DeAngelis (2015) for additional details. I thank Kris Allee and Matt DeAngelis for their help in understanding the ARF calculations. I also thank Petr Savický for posting his Perl script for computing ARF as described in Savický and Hlavácová (2002). My ARF implementation is based heavily on his script.
Δ*GROUP*	*GROUP* from the postperiod 10-K footnotes minus *GROUP* from the preperiod 10-K footnotes.
TABNUMS	The percentage of *NUMS* related to tabular presentation. In other words, it is the count from the second procedure described for the *NUMS* variable, which is then multiplied by 100 and divided by the total count of DH numbers.
Δ*TABNUMS*	*TABNUMS* from the postperiod 10-K footnotes minus *TABNUMS* from the preperiod 10-K footnotes.
	Partitioning variable and control variables
DHFIRM	An indicator variable set equal to 1 if the firm mentions SFAS 161 in any of its 10-K or 10-Q filings (or variants of those filings) filed between March 1, 2008, and December 31, 2009, and also has at least 10 DH words or phrases in its pre- and post-SFAS 161 10-K filing footnotes. To identify SFAS 161 terminology, I used the EDGAR full-text search to search for "161" in 51 10-K forms and 50 10-Q forms filed between October 16, 2009, and October 31, 2010 (at the time the data were collected, the earliest available date for the full-text search was October 16, 2009). I noted the various ways in which firms referred to SFAS 161. Examples include "SFAS No. 161" and "Statement of Financial Accounting Standards No. 161." I then converted these various phrases into regular expressions and used a Perl script to strip all filings of HTML and search for these phrases. The regular expressions are as follows: "\(?.?S?FAS.?\)? (No\.)? ?161"; "Standards (No\.)? ?161"; "Statement (No\.)? ?161"; "FAS-161"; "\(?.?ASC.?\)? ?815"; "ASC Topic 815"; and "ASC section 815".
POST	An indicator variable set equal to 1 (0) for all observations on or after (before) the filing date of the firm's initial SFAS 161 filing. Note that the adoption date used to determine the value of *POST* can be different for each firm. In other words, there is not one date applied to all firms in determining *POST*. The adoption date is determined by filings meeting one of the following two criteria: (1) a first, second, or third fiscal quarter ending between February 15, 2009, and May 15, 2009, or (2) a first fiscal quarter ending between May 16, 2009, and August 15, 2009.
DRISK	The weighted average of daily absolute percentage changes in a firm's underlying risk factors. The 17 risk factors used are interest rates on 1-year, 3-year, 5-year, and 10-year interest rate swaps (*dswp1*, *dswp3*, *dswp5*, and *dswp10*, respectively, from the Interest Rate Swap data available from the Federal Reserve Economic Data (FRED) maintained by the Federal Reserve Bank of St. Louis); U.S. dollar exchange rates for the euro, Japanese yen, pound sterling, Australian dollar, Canadian dollar, Swiss franc, Mexican peso, and Chinese yuan (*exuseu*, *exjpus*, *exusuk*, *exusal*, *excaus*, *exszus*, *exmxus*, and *exchus* variables, respectively, from the daily Federal Reserve Foreign Exchange Rates data set on WRDS); and the following S&P GSCI Indices: Energy, Agricultural, Industrial Metals, Livestock, and Precious Metals (*SPGSEN*, *SPGCAG*, *SPGSIN*, *SPGSLV*, and *SPGSPM* Bloomberg tickers, respectively). Using different swap maturities allows for differences in the length of firms' interest rate exposures. I use the eight most common currencies for U.S. dollar-paired foreign exchange deals (see table 3 in the BIS 2013 foreign exchange report: https://www.bis.org/publ/rpfx13fx.pdf). Requiring sample firms to have at least 15 monthly returns between January 1, 2005, and December 31, 2007, I run 17 regressions for each firm in order to estimate its exposure to each of the 17 risk factors. I then use the absolute coefficient estimates to divide the firms into separate terciles for each of the 17 risk factors; this means a firm could be in the bottom tercile for one factor and the top or

Appendix B. (Continued)

Variable	Definition
	middle tercile for another. Firms in the bottom tercile for a particular factor are assigned a weight of 0 for that factor, representing little to no exposure. Similarly, firms in the middle (top) tercile for a particular factor are assigned a weight of 1 (2) for that factor, capturing medium (high) exposure. I then calculate the daily absolute percentage change for each of the 17 risk factors and calculate the weighted average for each firm, using each firm's specific weights. This procedure transforms 17 daily marketwide movements into one summary measure of movement in risk factors that is specific to each firm-day in the sample. See Section IA.3 of the internet appendix for more details.
ROA	Return on assets measured from the Compustat Quarterly database as *NIQ* scaled by *ATQ*.
LEV	Leverage measured from the Compustat Quarterly database as *LTQ* scaled by *ATQ*.
BTM	Book-to-market ratio measured from the Compustat Quarterly database as *SEQQ* scaled by (*PRCCQ* × *CSHOQ*).
SIZE	Firm size measured from the Compustat Quarterly database as the natural log of *ATQ*.
PRICE	The average price during the calendar month following the 10-Q or 10-K filing date. For each firm-day, the price is measured from the CRSP Daily Stock File as the absolute value of *PRC*.
TURNOVER	The natural log of the average daily trading volume during the calendar month following the 10-Q or 10-K filing date. For each firm-day, the volume is measured from the CRSP Daily Stock File as *VOL* scaled by (*SHROUT* × 1,000).
FOLLOW	The natural log of 1 + the number of unique analysts issuing an earnings per share (EPS) forecast during the three calendar months ending with the 10-Q or 10-K filing date. Unique analysts are identified by the *ANALYS* variable in the Institutional Brokers' Estimate System (IBES) Detail U.S. EPS file.
VIX	The average of the CBOE Volatility Index during the calendar month following the 10-Q or 10-K filing date.
PASTSTDRET	The standard deviation of daily returns (CRSP Daily Stock File *RET*) during the calendar month ending one day prior to the 10-Q or 10-K filing date.
TRADESIZE	The average trade size during the calendar month following the 10-Q or 10-K filing date. Average trade size is obtained from the TAQ-based WRDS Intraday Indicators database by dividing the total volume by the total number of trades occurring each day.
	Real effects variables
Δ*RISKEXP*	The difference between $RISKEXP_{POST}$ and $RISKEXP_{PRE}$. $RISKEXP_{PRE}$ ($RISKEXP_{POST}$) is a measure of each firm's overall risk exposure before (after) SFAS 161. These measures are constructed in the same way as the denominator of Equation (IA.1) in the internet appendix, except that instead of using data from 2005 to 2007, I use data from 2007 to 2008 for $RISKEXP_{PRE}$ and 2009 to 2010 for $RISKEXP_{POST}$. $\lvert \Delta RISKEXP \rvert$ is the absolute value of Δ*RISKEXP*.
Δ*10KSIZE*	The percentage change in the Loughran and McDonald 10-K "net file size" to capture potential changes in overall disclosure (see https://sraf.nd.edu/data/, data downloaded December 30, 2016).
Δ*LINECREDIT*	The change in the percentage of 10-K Management Discussion and Analysis (MD&A) and footnote words related to lines of credit to capture changes in the firm's reliance on lines of credit. The phrases used follow Berrospide and Meisenzahl (2015) and include the following: "credit facility(ies)," "credit line(s)," "line(s) of credit," "loan facility(ies)," "revolving facility(ies)," and "term loan(s)."
Δ*CRISIS*	The change in the percentage of 10-K MD&A and footnote words related to the financial crisis. The words and phrases are the following: "crisis(es)," "recession(s)," "downturn(s)," "down turn(s)," "slowdown(s)," "slow down(s)," "subprime(s)," "sub prime(s)," "nonprime(s)," "non prime(s)," "meltdown(s)," "melt down(s)," "contraction(s)," "disruption(s)," "disrupt(ed)," "deterioration(s)," "deteriorate(d)(s)," "credit loss(es)," "tight(ened)(ening) credit," "reduced(ing) credit." To determine these words and phrases, I used SeekEdgar to anecdotally examine 10-K forms from 2008 and 2009. I searched for "crisis" and looked for other words and phrases commonly used when discussing the financial crisis.
Δ*NOTESPAY*	The change in the annual amount of notes payable (scaled by total assets) from pre- to post-SFAS 161 adoption to capture changes in the firm's reliance on short-term notes payable.
Δ*CRATING*	An indicator variable equal to 0 if there is no change in the investment-grade status of the firm's long-term credit rating from S&P and equal to 1 (−1) if the firm has at least one noninvestment grade rating in the post-SFAS 161 (pre-SFAS 161) period and zero noninvestment grade ratings in the pre-SFAS 161 (post-SFAS 161 period).
Δ*CASH*	The change the average level of cash and short-term investments (scaled by total assets) over each firm's four pre-SFAS 161 quarters to its four post-SFAS 161 quarters to capture changes in the firm's use of cash.

Endnotes

[1] I summarize the assumptions here; Sections 2.3 and 3.1 provide more details in the context of SFAS 161.

[2] As explained in Section 4.3.1, I find empirical support for this assumption by showing that fluctuations in firms' market risk factors (e.g., commodity prices or interest rates) are positively associated with bid-ask spreads.

[3] In Sections 2 and 3, I describe the content and formatting requirements of SFAS 161 and how my textual measures correspond to them. I view content as the "substance" of the disclosure and format as the "arrangement" of the disclosure (see definitions for "content" and "format" in the *New Oxford American Dictionary*, 3rd ed. Edited by Angus Stevenson and Christine A. Lindberg. New York: Oxford University Press, 2010. https://doi.org/10.1093/acref/9780195392883.001.0001).

[4] In the pre-SFAS 161 period, Compustat's DH data coverage is limited to this particular type of DH activity.

[5] FASB Statement No. 133, "Accounting for Derivative Instruments and Hedging Activities," was issued in June 1998 and was effective for all fiscal quarters beginning after June 15, 2000. Statement No. 161, "Disclosures about Derivative Instruments and Hedging Activities," was issued in March 2008 and was effective for fiscal years and interim periods beginning after November 15, 2008. The accounting and disclosure requirements for DH activities are now part of Accounting Standards Codification Topic 815. The SEC's Financial Reporting Release No. 48 (FRR 48) also requires firms to provide information about derivatives and risk exposures (Roulstone 1999, Linsmeier et al. 2002). These disclosures are not part of the audited financial statement footnotes affected by SFAS 161.

[6] The remainder of this section discusses the *changes* caused by SFAS 161; the tabular comparison in Appendix A is structured to facilitate a direct comparison of disclosure requirements under SFAS 133 and SFAS 161.

[7] Hedging instruments are those that (1) qualify for hedge accounting treatment under SFAS 133 and (2) are designated as hedging instruments by management. If an instrument qualifies for hedge accounting treatment, management chooses whether to designate it as a hedging instrument.

[8] An exception is Wong (2000), who finds weak and mixed evidence of SFAS 119 derivative disclosures' usefulness for assessing firms' risk exposures. Another stream of research studies the effects of risk-related disclosures not specific to derivatives and hedging. These papers find that risk disclosures are useful sources of information and are associated with investor perceptions of firm risk (Kravet and Muslu 2013, Campbell et al. 2014, Bao and Datta 2014, Filzen 2015, Hope et al. 2016).

[9] Chen et al. (2021) and Campbell et al. (2021) study larger samples in the specific contexts of customer-supplier relationships and cash flow hedges.

[10] By focusing on the information asymmetry effects of SFAS 161, I assume that SFAS 161 impacted DH disclosures but did not lead to confounding shifts in firms' underlying DH activities. However, because it is possible that firms change their DH activities because of the financial crisis and/or SFAS 161 (Chiorean 2016, Calluzzo and Dudley 2018), my empirical analyses are explicitly designed to mitigate these concerns (see Section 4.3.4). In addition, I focus on information asymmetry and not firm value because "an identical asset held or a liability owed by two different entities can have very different implications" for those entities' intrinsic values (FASB 2014, paragraph D14). Thus, it is not possible to predict how more disclosure about firms' DH activities would affect firm value, even if firms are exposed to similar risks. An alternative perspective would be to consider the impact of expanded DH disclosures in option markets, where risk information is directly related to valuation (e.g., Cherian and Jarrow 1998, Nandi 2000, and Smith 2019). However, because option markets may attract a more sophisticated subset of investors (e.g., Pan and Poteshman 2006, Johnson and So 2012, and Akbas et al. 2016), I focus on information asymmetry in equity markets to speak more directly to SFAS 161's objective of helping all types of investors better understand the firm-value implications of firms' DH activities.

[11] His remarks come from the October 17, 2016, FASB webinar, *IN FOCUS: FASB's Proposed Accounting Standards Update on Hedging*.

[12] Focusing on the more specific context of cash flow hedges, Campbell et al. (2021) conclude that SFAS 161 disclosures help analysts "better incorporate the effect of unrealized cash flow hedge gains and losses on future profitability into their earnings forecasts" (p. 171).

[13] Consistent with these mixed views, prior theoretical research suggests that the impact of risk-factor-related disclosure on the cost of capital can be ambiguous (Heinle and Smith 2017, Heinle et al. 2018).

[14] *POST* is included as a main effect in Equation (1) because the initial SFAS 161 filing date is specific for each firm; thus, *POST* is not entirely subsumed by the calendar year-quarter fixed effects (*TIME*). This approach allows me to take advantage of staggered adoption (to the degree it exists among firms adopting around the same time).

[15] My results provide support for this assumption and are explained in Section 4.3.1.

[16] In addition to spreads, Bens et al. (2016) also consider return volatility over the month following firms' 10-K filings. They view both the bid-ask spread and return volatility as measures of "information uncertainty." I do not focus on return volatility because it is not clearly related to the FASB's stated goals for issuing SFAS 161.

[17] My main tests focus on average bid-ask spreads during the month following 10-K/10-Q filings (Bens et al. 2016), but DH disclosures should also be useful to investors interpreting the firm-value implications of DH activities in the context of market risk factor fluctuations at any point in time. As a result, I also test whether SFAS 161 affects the empirical association between *daily* bid-ask spreads and *daily* fluctuations in firms' underlying risk factors (e.g., daily changes in oil prices) at all times—not just near 10-K/10-Q filings. See Section IA.3, Table IA.3, and Table IA.4 in the internet appendix for more discussion of these results, which provide additional support for the main findings.

[18] An alternative approach would be to count the number of individual footnotes containing DH information and check for cross-references. However, this approach is difficult to automate in a large sample because of considerable heterogeneity in the way firms format and label individual footnotes.

[19] SeekEdgar provided the service of separating the financial statement footnotes from all sample firms' pre- and post-SFAS 161 10-K filings. SFAS 161 disclosure requirements are not different for annual and interim periods. However, I do not use initial 10-Q filings to calculate the disclosure characteristic variables because only 10-K filings are required to contain DH footnote disclosures in the pre-period. To keep the basis for disclosure comparison constant from the pre- and post-periods, I use the 10-K from each period to calculate the changes in disclosure characteristics. Appendix B provides additional details for each disclosure characteristic variable.

[20] For each disclosure change variable except $\Delta GROUP$, more positive values indicate a stronger impact of SFAS 161 on DH disclosure. However, more positive values of $\Delta GROUP$ indicate the opposite (more dispersed information). To maintain interpretations across the four disclosure change measures, $\Delta GROUP$ is multiplied by −1 for all analyses and tables, including the factor analysis described later.

[21] *DHDISC* is not a differenced variable like the other disclosure change measures because the inputs to the factor analysis *are the disclosure change measures*. The factor analysis yields one factor with an eigenvalue greater than 1 (2.032). Regarding the factor loadings, $\Delta WORDS$ and $\Delta NUMS$ load strongest (loadings of 0.8567 and 0.8071, respectively), followed by $\Delta TABNUMS$ (loading of 0.6575) and $\Delta GROUP$ (loading of 0.4632). It should also be emphasized that $\Delta WORDS$, $\Delta NUMS$, $\Delta GROUP$, $\Delta TABNUMS$, and *DHDISC* are *firm-level variables* that do not vary with time during the sample period.

[22] Although this second weakness would work against finding results, it is still problematic from the perspective of accurately identifying the firms that were less versus more strongly affected by SFAS 161.

[23] In Section 4.3.4, I discuss additional analyses that alleviate these concerns.

[24] This procedure ignores early adoption; however, SFAS 161 was issued in March 2008, leaving little time for early adoption before the November 15, 2008 effective date. In addition, an anecdotal examination of filings from this period indicated that few firms appeared to adopt SFAS 161 before they were required to do so.

[25] As described in more detail in Appendix B, DH firms are identified as those mentioning SFAS 161 in 10-Q or 10-K filings filed between March 1, 2008, and December 31, 2009, and having at least 10 DH words, phrases, or acronyms in their pre- and post-SFAS 161 10-K footnotes. Inspection of this categorization procedure indicates that firms not mentioning SFAS 161 do not use derivatives and are thus not affected by SFAS 161. However, some nonderivative users mention SFAS 161; anecdotally, this occurs when a firm describes all new accounting standards in its SEC filings, even if certain standards do not affect the firm. Under my procedure, these firms would be misclassified as DH firms. The requirement for at least 10 DH dictionary items is meant to mitigate this risk of misclassification. As previously explained, *DHACTIVITY* does not capture all types of DH activities, which is why I do not rely on *DHACTIVITY* to classify DH versus non-DH firms. In Section 4.3.1, I describe robustness tests using matching procedures based on alternative methods of classifying DH versus non-DH firms.

[26] These three industries are finance (FF-11), with 22.9% of the sample (e.g., Bank of America and Citigroup); business equipment (FF-6), with 17.9% (e.g., IBM and Apple); and other (FF-12), with 11.6% (e.g., General Electric and UPS).

[27] The internet appendix contains additional information about the treatment measures shown in Table 2. Figure IA.2 presents time series trends in the levels of each of the disclosure content and format measures surrounding the adoption of SFAS 161. Table IA.2 provides correlations and industry-level statistics for the treatment measures among DH firms. The accompanying discussion of this figure and table appears in Section IA.1 of the internet appendix.

[28] I winsorize all continuous variables in Table 3 at 1% and 99% to address potential outliers. Bens et al. (2016) do not mention winsorization, but I opt to winsorize because the resulting descriptive statistics are closer to their reported statistics.

[29] For the regressions, *DHDISC* and *DHACTIVITY* have been scaled by their sample standard deviation among DH firms to facilitate easier comparison of the effect sizes across models. Although I am primarily interested in the sign of the *POST* × *DHDISC* and *POST* × *DHACTIVITY* coefficients, comparing their magnitudes allows for evaluating the consistency of the results across the two treatment measures.

[30] In addition to using *DHFIRM* to identify treated and control firms for the coarsened exact matching and entropy balancing, I also test two alternative classification methods where (1) treated (control) firms are those with values of *DHDISC* above (below) the sample median, and (2) treated (control) firms are those with *DHACTIVITY* greater than (equal to) 0. For coarsened exact matching, I also tested results using two and four bins for the coarsening of the preadoption averages. For entropy balancing, I also tested results matching on (1) only the mean and (2) the mean, variance, and skewness. The (untabulated) results are similar across all these various specifications.

[31] Because $DHACTIVITY = 0$ for 63% of the sample, I group these observations together in Panel B and label them $DHACTIVITY_{\text{Quintile1 to Quintile3}}$, and the remaining observations with nonzero *DHACTIVITY* are divided into two groups with each having roughly the same number of observations as each *DHDISC* quintile.

[32] When considering the results in Table 6, Panel B, it is important to note that the means of the 10,000 bootstrap coefficients are expected to move *closer to* 0 (relative to the coefficients in Table 4), but they are *not* expected to be *equal to* 0. With only eight quarters of data per firm (i.e., four preadoption and four postadoption), there is a mechanical limit on the degree to which *POST* can be altered for each firm (on average, the redefined values of *POST* match the true values of *POST* for six of a firm's eight sample observations). In other words, although the bootstrap procedure introduces noise into the redefined *POST* variable, the sample composition prevents the bootstrapping from introducing enough noise to drive the bootstrap coefficients to 0.

[33] The sample size is reduced in Tables 7 and 8 from 2,079 firms to 2,071 firms as a result of missing values of some real effects variables.

[34] The *POST* × *DHDISC* and *POST* × *DHACTIVITY* coefficients also remain very similar to those shown in Table 4 if the real effects controls are considered separately rather than simultaneously. Section IA.2 of the internet appendix describes additional robustness checks related to real effects.

[35] For ease of interpretation across measures, in the regression models, all disclosure-based *TREAT* measures have been scaled by the corresponding measure's sample standard deviation among DH firms.

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